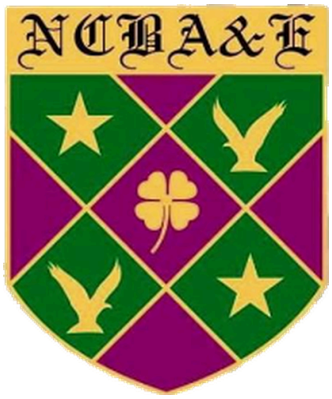


NATIONAL COLLEGE OF BUSINESS ADMINISTRATION & ECONOMICS

Front Lane Campus, Lahore

FINAL YEAR PROJECT REPORT

SmartSplit & ExpenseAI

An AI-Assisted Personal Finance Manager and Group Expense Settlement System using Min-Cash-Flow Optimisation

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A report submitted in partial fulfilment of the requirements for the award of the Bachelor of Science degree (BSCS and BSITM programmes)

Department of Computer Science

National College of Business Administration & Economics, Front Lane Campus, Lahore

Session [Session] · October 2026

DECLARATION

We, **Muhammad Azeem** (Roll No. BSCSvb-f24152), **Chanda Safdar** (Roll No. BSCS-vb-f24-133) and **Saba** (Roll No. BSITM-f24-25), hereby declare that this project report titled "*SmartSplit & ExpenseAI: An AI-Assisted Personal Finance Manager and Group Expense Settlement System using Min-Cash-Flow Optimisation*" is our own work and has not been submitted, in whole or in part, for any other degree or qualification at this or any other institution. All sources of information used in this report have been duly acknowledged and referenced.

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CERTIFICATE OF APPROVAL

It is certified that the project report titled "*SmartSplit & ExpenseAI: An AI-Assisted Personal Finance Manager and Group Expense Settlement System using Min-Cash-Flow Optimisation*" submitted by **Muhammad Azeem** (Roll No. BSCSvb-f24152), **Chanda Safdar** (Roll No. BSCS-vb-f24-133) and **Saba** (Roll No. BSITM-f24-25) has been examined and found satisfactory in partial fulfilment of the requirements for the award of the Bachelor of Science degree (BSCS and BSITM programmes) at the Department of Computer Science, National College of Business Administration & Economics, Front Lane Campus, Lahore.

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ABSTRACT

Managing money is a daily problem for students, young professionals and households in Pakistan. Personal spending is usually tracked in notebooks or not at all, and shared costs on trips or in shared flats lead to long chains of small, confusing repayments. Widely used tools are either designed for foreign markets, keep key features behind paid plans, or treat personal finance and group splitting as two separate products.

This project presents **SmartSplit & ExpenseAI**, a full-stack web application that combines both needs in one account. The personal workspace records income and expenses in Pakistani rupees, applies monthly category budgets, generates recurring bills automatically, and reads printed receipts using a vision language model, with Tesseract.js as an offline fallback. Every extracted line is verified by the user in a human-in-the-loop dialog before it is saved, and verified items are compared against a benchmark of 100 grocery products from seven Pakistani stores to suggest cheaper alternatives. Savings goals show how much has been put aside, whether the user is on pace, and how much to save each month. The group workspace supports multi-payer expenses, four split methods (equal, exact, percentage and line item), expenses entered in foreign currencies on trips abroad and converted to rupees at daily exchange rates, and a settlement engine based on the greedy Min-Cash-Flow algorithm, which guarantees that a group of V members is settled in at most $V - 1$ payments.

The system is built on the MERN stack (MongoDB, Express, React and Node.js) and exposes 79 REST endpoints. Money is stored as integer rupees, balances are always recomputed from expenses by database aggregation, and every group request is authorised on the server. Authentication uses short-lived JSON Web Tokens with rotating refresh tokens and Google OAuth 2.0. The implementation is verified by 288 automated unit, API and property-based tests, all passing, and is deployed publicly with every change tested before it goes live. On the demonstration trip of five members and seven expenses, the settlement engine reduced 20 naive pairwise debts to 4 payments.

Keywords: expense management, debt simplification, Min-Cash-Flow, greedy algorithm, receipt OCR, vision language model, multi-currency, savings goals, MERN stack, JWT

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
CI/CD	Continuous Integration / Continuous Deployment
CORS	Cross-Origin Resource Sharing
CSV	Comma-Separated Values
ERD	Entity Relationship Diagram
FR	Functional Requirement
FX	Foreign Exchange
FYP	Final Year Project
HTTP(S)	Hypertext Transfer Protocol (Secure)
JSON	JavaScript Object Notation
JWT	JSON Web Token
LLM	Large Language Model
MCF	Min-Cash-Flow
MERN	MongoDB, Express, React, Node.js
NFR	Non-Functional Requirement
OAuth	Open Authorization
OCR	Optical Character Recognition
ODM	Object Document Mapper
OWASP	Open Worldwide Application Security Project
PDF	Portable Document Format
PKR	Pakistani Rupee
REST	Representational State Transfer
SMTP	Simple Mail Transfer Protocol
SPA	Single-Page Application
SSO	Single Sign-On
TTL	Time To Live
UI	User Interface
XLSX	Office Open XML Spreadsheet
XSS	Cross-Site Scripting

CHAPTER 1
INTRODUCTION

This chapter introduces the problem domain, states the problem addressed by the project, and sets out its objectives, scope, significance and development methodology. It closes with an outline of the remaining chapters.

1.1 Background

Personal financial management is the routine of recording income and expenditure, comparing it with a plan, and adjusting behaviour. Research and everyday experience agree that people who track their spending make better financial decisions, yet most people find manual tracking tedious and stop after a few weeks. Receipts are lost, small cash purchases are forgotten, and categories are applied inconsistently.

A second, closely related problem appears whenever costs are shared. University trips, shared flats, weddings and office events produce many bills paid by different people for different subsets of the group. At the end, each participant needs to know how much they owe or are owed, and the group needs a practical way to settle. Settling every bill separately produces many small transfers; with mobile wallets such as JazzCash and Easypaisa and the State Bank of Pakistan's instant payment system Raast [19], each transfer is easy, but the number of transfers and the confusion about who has already paid remain the real cost.

Recent advances make a better solution practical. Vision language models can read photographs of printed receipts and return structured data, and established algorithms can reduce a web of debts to a small set of payments [1]. Modern web frameworks allow both capabilities to be delivered in a responsive application that works on any phone or laptop browser without installation.

1.2 Problem Statement

Existing tools address personal tracking and group splitting separately, require manual entry of every item, rarely support Pakistani payment methods or price references, and do not explain how settlement payments are derived. As a result, users must maintain two systems, re-type receipt contents by hand, and settle group debts through more transfers than necessary.

The project therefore addresses the following question: *how can a single, secure web application help users record personal and shared expenses with minimal manual effort, guarantee that every rupee is accounted for exactly, and settle group debts with the fewest practical number of payments?*

1.3 Objectives

The objectives of the project are to:

1. Design and implement a personal workspace for recording income and expenses in PKR with categories, local payment methods, monthly budgets and automatically generated recurring entries.
2. Implement AI-assisted receipt scanning that extracts line items from a photo using a vision language model, falls back to offline OCR, and requires human verification before saving.
3. Suggest savings by matching verified receipt items against a local benchmark of market prices.
4. Implement a group workspace with role-based membership, multi-payer expenses and four split methods that always sum exactly to the bill total.
5. Implement a settlement engine based on the greedy Min-Cash-Flow algorithm that settles a group of V members in at most $V - 1$ payments, and demonstrate the reduction compared with naive repayment.
6. Help users save towards personal goals, and let groups record trips abroad in foreign currencies while all accounting stays in rupees.
7. Provide reports, charts and CSV, Excel and PDF exports for both workspaces.
8. Secure the system with industry-standard authentication, authorisation, input validation and rate limiting.
9. Verify correctness through automated unit, API and property-based tests.
10. Deploy the system publicly so that every change is tested automatically before it goes live.

1.4 Scope

In scope. A responsive web application for individual users and small groups (trips, flats and events of roughly 2 to 20 members); amounts in Pakistani rupees, with group expenses also accepted in 15 foreign currencies and converted to rupees; savings goals; email/password and Google sign-in; receipt photos in JPG or PNG format up to 8 MB; price benchmarks for about 100 grocery products from seven Pakistani stores; deployment on free cloud tiers.

Out of scope. Moving money between accounts (users pay through their own wallet or bank and then mark the payment as paid), bank account synchronisation, balances kept in a currency other than the rupee, native mobile applications, live web scraping of prices, and tax or investment features.

1.5 Significance of the Project

- **For users:** one account covers personal and shared money; receipts are captured in seconds; groups settle with the minimum practical number of transfers.
- **For the local context:** PKR formatting, JazzCash, Easypaisa and Raast as first-class payment methods, and prices from Pakistani retailers.
- **Academically:** the project applies a graph-based greedy algorithm with a provable bound, integer arithmetic for financial correctness, human-in-the-loop AI, fuzzy string matching and secure token-based authentication in a single coherent system.

1.6 Development Methodology

The system was developed using the **incremental model** [20]. The work was divided into ten increments (Phases 0 to 9), each delivering a usable slice of functionality. Every increment followed the same cycle: refine the requirements, implement, write automated tests, and record a phase report before the next increment began. Core money logic (split calculation and settlement) was written test-first. Figure 1.1 shows the increments.

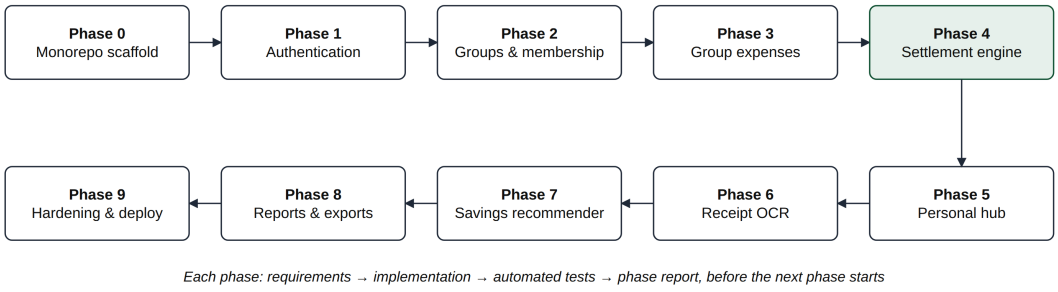


Figure 1.1: Incremental development phases of the project

The incremental model was preferred over the waterfall model because later features (OCR, savings suggestions and reports) depend on earlier ones (authentication, groups and expenses), and delivering each layer fully tested reduced the risk of defects compounding across the system. After the ten planned increments, further increments based on user and supervisor feedback added an invitation inbox, member roles, a price book with super-administrators, public deployment, multi-currency trip expenses and savings goals.

1.7 Organisation of the Report

Chapter 2 reviews existing systems and the relevant literature. Chapter 3 specifies the requirements. Chapter 4 presents the system design, including architecture, database and algorithms. Chapter 5 describes the implementation and user interface. Chapter 6 reports testing and evaluation. Chapter 7 concludes and outlines future work. Appendices provide the API reference, a user manual and an installation guide.

CHAPTER 2
LITERATURE REVIEW

This chapter reviews existing applications for personal and group expense management, the algorithmic background of debt simplification, techniques for receipt text extraction and product matching, and the security standards used by modern web applications. It concludes by identifying the gap addressed by this project.

2.1 Personal Finance Management Applications

Personal finance applications such as Wallet by BudgetBakers [17] and many similar budgeting apps allow users to record transactions, group them into categories and compare them against budgets. Spreadsheets remain the most common free alternative. These tools are effective for disciplined users, but entry is largely manual and they are not designed for splitting costs with other people. Support for Pakistani payment methods and local price references is limited.

2.2 Group Expense Splitting Applications

Splitwise [15] and Tricount [16] are widely used for shared expenses. Both record who paid for what, compute balances and propose repayments; Splitwise offers a "simplify debts" option that reduces the number of repayments. Advanced features such as itemised receipt entry are offered in paid tiers in some products. These applications focus on the group and do not provide a full personal budgeting workspace, so users who want both must use two separate systems.

2.3 Debt Simplification Algorithms

A set of shared expenses can be modelled as a directed weighted graph in which an edge from A to B of weight w means that A owes B an amount w. Only each person's **net balance** (total paid minus total owed) matters for settlement: any set of payments that brings every net balance to zero is a valid settlement. Verhoeff [1] analyses this problem in detail and shows that finding the absolute minimum number of payments is computationally hard, since it requires partitioning the members into the largest possible number of zero-sum subgroups, which is related to the NP-complete subset-sum problem [2].

The practical alternative is a **greedy** strategy: repeatedly match the largest creditor with the largest debtor and transfer the smaller of the two amounts. Each transfer clears at least one person, so a group of V members needs at most $V - 1$ payments [1], [3]. This bound is tight

in general and the method runs in polynomial time, which makes it the standard choice in practice. This project adopts the greedy Min-Cash-Flow method and reports its result against the naive pairwise approach.

2.4 Receipt Text Extraction (OCR)

Traditional OCR engines such as Tesseract [4] segment an image into lines and characters and recognise them using trained models. They work offline and at no cost, but return unstructured text; a separate rule-based step is needed to identify item names, quantities and prices, and accuracy drops on crumpled, faded or photographed receipts. Vision language models accept an image directly and can be instructed to return a fixed JSON structure, which removes the need for hand-written parsing rules and handles varied layouts better [14]. Because any automated extraction can misread digits, financial applications should keep a human in the loop to confirm results before they affect stored balances.

2.5 String Similarity for Product Matching

Receipt item names are abbreviated and inconsistent ("COOKING OIL 3LTR", "Cooking Oil 3 Litre"). The Sørensen–Dice coefficient [5] measures the similarity of two strings as twice the number of shared character bigrams divided by the total number of bigrams, giving a score between 0 and 1. It is simple, fast and robust to small differences in word order and spelling, which makes it suitable for matching receipt text against a product catalogue after normalisation.

2.6 Web Application Security

The OWASP Top 10 [9] lists the most critical web application risks, including broken access control, cryptographic failures, injection and identification and authentication failures. JSON Web Tokens [6] allow a server to issue signed, self-contained access tokens; keeping them short-lived and pairing them with rotating refresh tokens limits the damage of a stolen token. OAuth 2.0 [7] enables sign-in through an identity provider such as Google without the application handling the user's password. Passwords that are stored must be hashed with a deliberately slow, salted algorithm such as bcrypt [8]. For financial values, binary floating-point arithmetic introduces representation errors [10], so amounts should be stored as integers of the smallest currency unit.

2.7 Comparison with Existing Systems

Table 2.1 compares the proposed system with representative existing tools, based on their publicly documented features at the time of writing.

Table 2.1: Comparison of the proposed system with existing tools

Feature	Spreadsheet	Wallet	Splitwise	Tricount	Proposed
Personal income/expense tracking	Manual	Yes	Limited	No	Yes
Monthly category budgets	Manual	Yes	No	No	Yes
Group expense splitting	Manual	No	Yes	Yes	Yes
Debt simplification	No	No	Yes	Yes	Yes, with $V-1$ bound shown
Itemised receipt OCR	No	Limited	Paid tier	Limited	Yes, human-verified
Price comparison / savings tips	No	No	No	No	Yes
JazzCash / Easypaisa / Raast	Manual	Limited	No	No	Yes
Personal and group in one account	Manual	No	No	No	Yes

2.8 Research Gap

The review shows that no single, free tool combines personal budgeting, AI-assisted and human-verified receipt capture, local price comparison and group settlement with a stated payment bound, for the Pakistani context. SmartSplit & ExpenseAI is designed to fill this gap, while applying established results from algorithms, OCR, string matching and security engineering.

CHAPTER 3
REQUIREMENTS ANALYSIS

This chapter identifies the users of the system and specifies its functional and non-functional requirements, use cases, business rules, and hardware and software requirements.

3.1 Users and Stakeholders

Table 3.1: User roles of the system

Role	Description
Guest	An unauthenticated visitor who can register, sign in (email or Google) and reset a forgotten password.
Registered user	Owns a private personal workspace (transactions, budgets, receipts, reports) and may belong to any number of groups.
Group member	An active member of a group who can view its expenses and balances, add expenses, preview a settlement and mark payments as paid.
Group admin	A member with management rights: edit group settings, add, invite or remove members, approve join requests, promote or demote members, rotate the share code and record settlements. The creator of a group is its first admin.
Super admin	A person whose email is listed in the ADMIN_EMAILS server setting. Sees the Admin page (totals, newest users and groups, service checklist) and maintains the benchmark price book.
External services	Vision AI (receipt extraction), Cloudinary (image storage), Google OAuth (sign-in), ExchangeRate-API (exchange rates), Brevo SMTP (email) and the nightly scheduler.

3.2 Functional Requirements

Table 3.2: Functional requirements

ID	Module	Requirement
FR-01	Accounts	The system shall allow users to register and sign in with email and password.
FR-02	Accounts	The system shall allow users to sign in with Google OAuth 2.0 and link it to an existing account with the same email.
FR-03	Accounts	The system shall provide password reset through a single-use emailed link that expires after 30 minutes.
FR-04	Accounts	The system shall allow users to update their profile and change their password.
FR-05	Personal	The system shall allow users to record money received (credit) and spent (debit) with title, amount, date, category, payment method and note.

ID	Module	Requirement
FR-06	Personal	The system shall provide seven built-in categories (Food, Utilities, Travel, Shopping, Healthcare, Housing, Misc) and allow user-defined categories.
FR-07	Personal	The system shall support the payment methods Cash, Bank Transfer, Card, JazzCash, Easypaisa and Raast.
FR-08	Personal	The system shall allow a monthly budget cap per category and show consumption as green (< 80%), amber (80–99%) or red (≥ 100%).
FR-09	Personal	The system shall generate weekly or monthly recurring transactions automatically, including any periods missed while the server was offline.
FR-10	Personal	The system shall show a monthly summary: received, spent, remaining, comparison with the previous month and spending per category.
FR-11	Receipts	The system shall accept a receipt photo (JPG/PNG, up to 8 MB) and process it in the background.
FR-12	Receipts	The system shall extract store, date, line items and total using a vision AI model, and fall back to offline OCR if the model is unavailable.
FR-13	Receipts	The system shall present extracted lines for user verification and editing, and warn when the lines and the printed total differ by more than 5%.
FR-14	Receipts	The system shall save a verified receipt as a personal transaction or as a pre-filled line-item group expense.
FR-15	Savings	The system shall compare verified items with benchmark prices and show cheaper alternatives and the total potential saving.
FR-16	Groups	The system shall allow users to create groups (trip, home, event, other) and become their admin.
FR-17	Groups	The system shall support joining a group by direct add, single-use email invite link, or share code with admin approval, and shall show pending invitations in an inbox where they can be accepted or declined.
FR-18	Groups	The system shall record group expenses paid by one or more members.
FR-19	Groups	The system shall split expenses equally, by exact amounts, by percentage, or by line item, with live validation messages.
FR-20	Groups	The system shall display each member's net balance, recomputed from all expenses and paid settlements.
FR-21	Settlement	The system shall preview the optimised payment list without saving, showing naive debt count versus optimised payment count.
FR-22	Settlement	The system shall let a group admin record a settlement, mark the group as settled, and keep an audit log entry.
FR-23	Settlement	The system shall allow each payment to be marked as paid, updating balances immediately.
FR-24	Reports	The system shall provide monthly and six-month personal reports and group summary reports with charts.
FR-25	Reports	The system shall export transactions and group reports as CSV, Excel (XLSX) and PDF.

ID	Module	Requirement
FR-26	Notifications	The system shall notify group members when an expense is added, a member joins, a settlement is created or a payment is marked paid.
FR-27	Groups	The system shall accept a group expense in any of 16 supported currencies, convert it to whole rupees at the current exchange rate, and keep the original amounts and rate for display.
FR-28	Goals	The system shall let users create savings goals with a target amount and date, record deposits and withdrawals, and show progress, whether they are on track and the amount to save each month.
FR-29	Groups	The system shall let invite links and share links be sent through WhatsApp, and let an admin promote or demote members while always keeping at least one admin.
FR-30	Admin	The system shall let super admins maintain the benchmark price book (add, edit, delete, Excel or CSV import and Excel export) and view an overview of users and groups.

3.3 Non-Functional Requirements

Table 3.3: Non-functional requirements

ID	Quality	Requirement
NFR-01	Correctness	All monetary values shall be integer rupees; every expense shall satisfy sum(splits) = total; group balances shall always sum to zero.
NFR-02	Security	Passwords shall be hashed with bcrypt (cost 12); access tokens shall expire in 15 minutes; refresh tokens shall rotate and be stored only as hashes in an HttpOnly cookie.
NFR-03	Security	Every group request shall verify active membership on the server; every request body shall be validated; authentication endpoints shall be rate limited.
NFR-04	Performance	Receipt upload shall respond immediately (HTTP 202) and process in the background; savings matching shall never block the user interface.
NFR-05	Reliability	OCR shall fall back to an offline engine; exchange rates shall fall back to the last saved copy; a failed notification or email shall never fail a money operation.
NFR-06	Usability	Every list shall have loading, empty and error states; forms shall validate inline with human-readable messages; amounts shall display as "Rs 4,200".
NFR-07	Portability	The application shall run in current desktop and mobile browsers and adapt to screens from 360 px wide.
NFR-08	Maintainability	Business logic shall live in pure, testable service modules; the API shall use consistent response envelopes.
NFR-09	Privacy	Receipt images shall be stored privately and served through signed URLs that expire after one hour.
NFR-10	Deployability	Every change pushed to the main branch shall be tested automatically and deployed only if all tests and the client build pass.

3.4 Use Case Diagrams

Figures 3.1 and 3.2 present the use cases of the account and personal workspace and of the group workspace respectively.

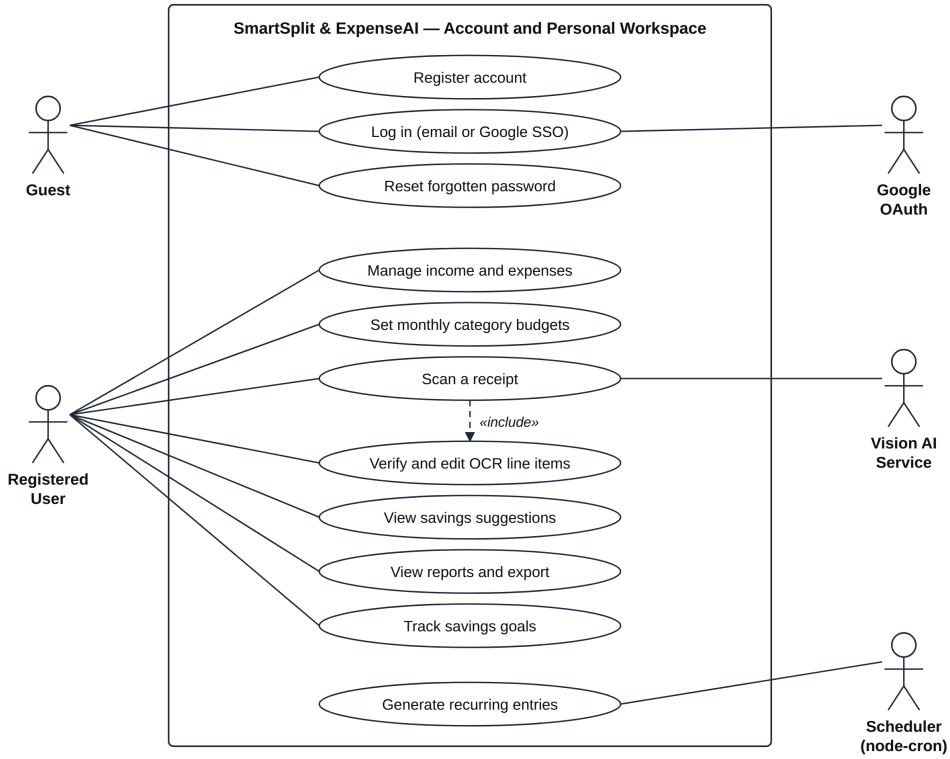


Figure 3.1: Use case diagram — account and personal workspace

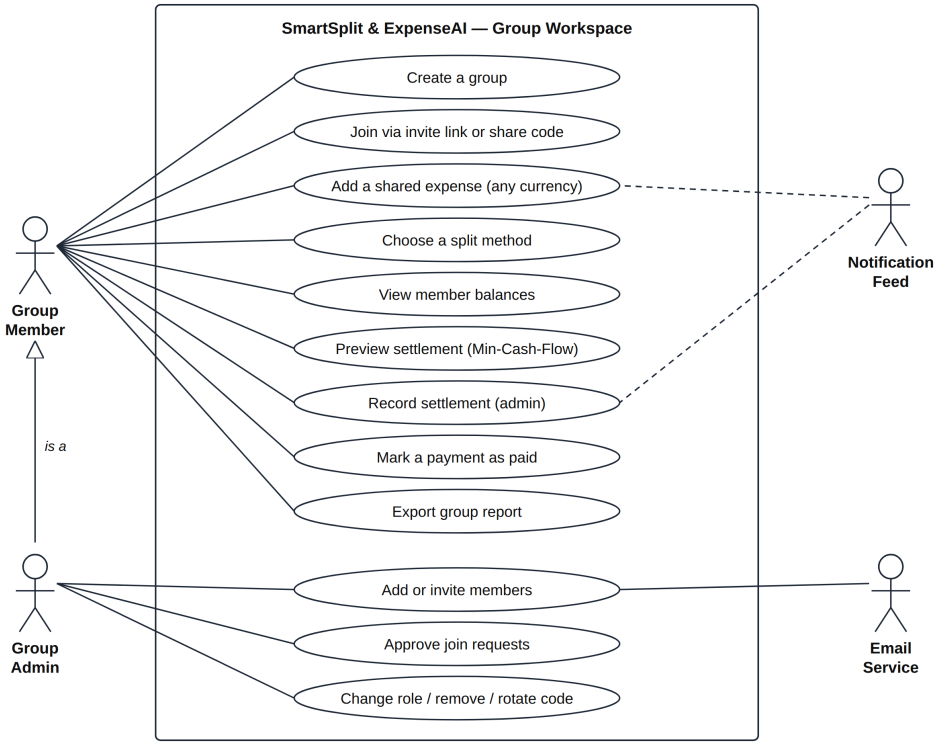


Figure 3.2: Use case diagram — group workspace

3.5 Use Case Descriptions

The three most important use cases are described below in tabular form.

Table 3.4: Use case UC-01: Add a shared expense

Use case ID	UC-01
Name	Add a shared expense
Actor	Group member
Precondition	The user is signed in and is an active member of the group.
Main flow	1. The user opens Add Expense and selects the group. 2. The user enters title, amount, date and category. 3. The user selects one or more payers and their amounts. 4. The user selects a split method and fills in the shares. 5. The system validates the input live and enables Save. 6. The server validates the input again, computes the splits, stores the expense and notifies other members.
Alternate flow	5a. Payers do not cover the total, exact amounts do not add up, or percentages do not total 100%: the system shows a message such as "You're Rs 50 short" and keeps Save disabled.
Postcondition	The expense is stored with sum(splits) = total, and the group balances reflect it.

Table 3.5: Use case UC-02: Settle up a group

Use case ID	UC-02
Name	Settle up a group
Actors	Group member (preview), group admin (record)
Precondition	The group has at least one non-zero balance and no unpaid settlement in progress.
Main flow	1. The user opens Settle Up. 2. The system shows the naive pairwise debts. 3. The user selects "Simplify with Min-Cash-Flow". 4. The system computes and displays the optimised payments and the number of transfers saved, without saving anything. 5. A group admin selects "Record settlement". 6. The system stores one settlement row per payment, marks the group as settled and writes an audit log entry.
Alternate flow	4a. Nobody owes anything: the system reports "nothing to settle". 6a. A settlement is already in progress: the system refuses with a conflict message.
Postcondition	Payments are recorded; when each is marked paid, the corresponding balances return to zero.

Table 3.6: Use case UC-03: Scan and verify a receipt

Use case ID	UC-03
Name	Scan and verify a receipt
Actors	Registered user, Vision AI service
Precondition	The user is signed in and has a photo of a printed receipt.
Main flow	1. The user uploads or captures a photo. 2. The system stores the image and replies immediately. 3. The vision AI extracts store, date, items and total in the background. 4. The system shows the verification dialog with the image and editable lines. 5. The user corrects, adds or removes lines and confirms. 6. The system saves a personal transaction (or prepares a line-item group expense) and later attaches savings suggestions.
Alternate flow	3a. The AI is unavailable: Tesseract.js extracts the lines offline. 4a. Lines and printed total differ by more than 5%: a warning shows both values.
Postcondition	Only user-verified data is stored.

3.6 Business Rules

Table 3.7: Business rules enforced by the system

ID	Rule
BR-01	Money is stored and calculated as integer rupees; fractional rupee amounts are rejected.
BR-02	Every expense satisfies sum(splits.share) = totalAmount and sum(paidBy.amount) = totalAmount.

ID	Rule
BR-03	Rounding remainders are assigned to the payer (the largest payer when there are several).
BR-04	Balances are always computed from expenses and paid settlements; they are never stored as running totals.
BR-05	Settlement uses greedy max-matching on net balances and produces at most V − 1 payments that bring all balances to zero.
BR-06	Every group route verifies, on the server, that the requesting user is an active member; client-sent identifiers are never trusted.
BR-07	OCR results are never saved automatically; the user verifies every line first.
BR-08	Price suggestions read the local benchmark collection first and never block the user interface.
BR-09	A foreign-currency expense is converted to whole rupees on the server; the rounding difference goes to the largest payer or row so that BR-02 still holds, and an edit keeps the original rate.
BR-10	Savings-goal progress is always computed from the recorded deposits and withdrawals, and a goal can never hold less than zero.
BR-11	Recording a settlement closes the group, so only a group admin may do it.

3.7 Hardware and Software Requirements

Table 3.8: Hardware and software requirements

Item	Development	End user
Processor / memory	Dual-core CPU, 8 GB RAM	Any device able to run a modern browser
Operating system	Linux, Windows or macOS	Any (Android, iOS, Windows, macOS, Linux)
Runtime	Node.js 20 or later, npm	Not required
Database	MongoDB 7 (local Docker) or MongoDB Atlas	Not required
Browser	Google Chrome (development and testing)	Chrome, Edge, Firefox or Safari (current versions)
Network	Internet for AI, image storage and Google sign-in	Internet connection
Tools	Visual Studio Code, Git, GitHub, Docker	None

CHAPTER 4
SYSTEM DESIGN

This chapter describes the architecture of the system, its database design, the algorithms at its core, the receipt processing pipeline, the security design and the API conventions.

4.1 System Architecture

The system follows a three-tier client–server architecture (Figure 4.1). The **presentation tier** is a React single-page application that runs in the browser. The **application tier** is a stateless REST API [11] built with Node.js and Express. The **data tier** consists of MongoDB and a set of external services. The browser never communicates with the database or the AI service directly: every request passes through the server's security checks.

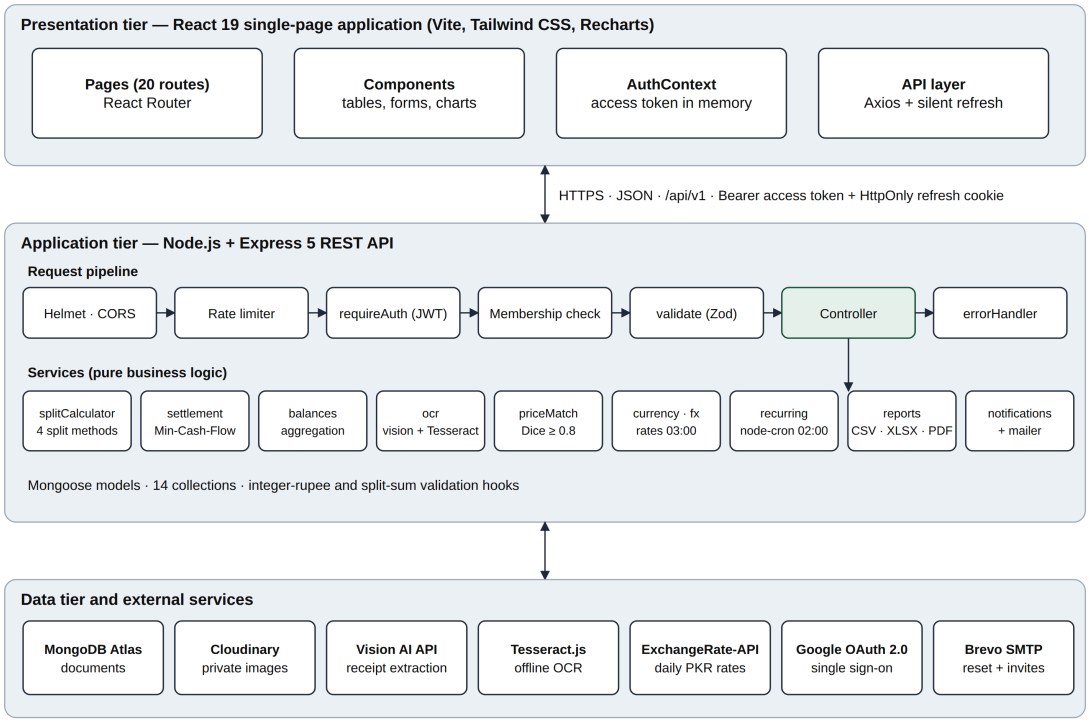


Figure 4.1: Three-tier architecture of SmartSplit & ExpenseAI

Each request passes through a fixed pipeline of middleware before it reaches a controller: security headers and CORS (Helmet), rate limiting, JWT authentication, group membership or admin checks, and Zod schema validation. Controllers are thin; the business logic lives in pure service modules (split calculation, settlement, balances, OCR, price matching, currency conversion and exchange rates, recurring entries, reports and notifications) that can be tested without a web server. Errors from any layer are converted by a single error handler into a consistent response envelope.

4.2 Database Design

MongoDB, a document database, was chosen because the central entity, a group expense, naturally contains nested lists: several payers, several split shares and several line items. Storing these inside one document means an expense is read and written atomically in a single operation. Mongoose [21] adds schemas, validation and indexes on top of MongoDB. Figure 4.2 shows the entity relationship diagram of the fourteen collections.

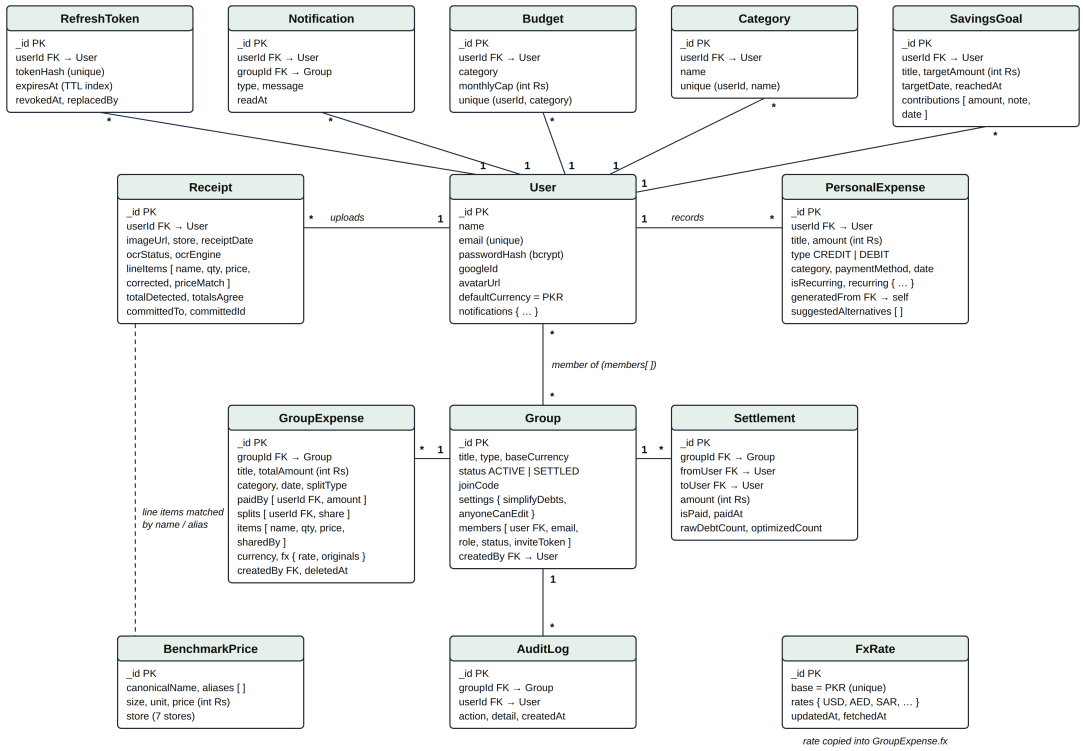


Figure 4.2: Entity relationship diagram

Table 4.1: Collections and their purpose

Collection	Purpose and key constraints
User	Account details; unique email; bcrypt password hash (never returned to the client); optional Google ID.
RefreshToken	One row per session; stores only the SHA-256 hash of the token; TTL index removes expired rows automatically.
Group	Group details and an embedded members array (role admin/member; status active/pending/requested); share code.
GroupExpense	Expense with paidBy, splits and items arrays; a validation hook rejects any document whose shares do not sum to the total; optional fx sub-document with the original currency, amounts and rate; soft delete via deletedAt.
Settlement	One planned payment (from, to, amount, paid flag) plus the naive and optimised payment counts.
AuditLog	Append-only record of important group actions such as settlement creation.

Collection	Purpose and key constraints
PersonalExpense	Personal credit or debit; recurring template settings; generatedFrom links generated copies to their template.
Budget	Monthly cap per category per user; unique on (userId, category).
Category	User-defined categories; unique per user, case-insensitive.
Receipt	Image reference, OCR status and engine, extracted line items with correction flags and price matches.
BenchmarkPrice	Product name, aliases, pack size and unit, price and store; 100 products and 609 price rows across seven stores.
Notification	One row per recipient with read state.
SavingsGoal	Goal title, target amount and date, and an embedded contributions array (positive deposits, negative withdrawals); reachedAt is set once.
FxRate	The latest rupee rates for the supported currencies with their date; a single row used as a fallback when the rate provider is unreachable.

Indexes support the most frequent queries: expenses by group and date, groups by member, transactions by user and date, due recurring templates by next run date, and unread notifications by user.

4.3 Expense Split Calculation

All split methods share one rounding rule so that every expense balances exactly. For a total T shared by n people, each person first receives the floor of T / n, and the remainder r is assigned to the payer:

$$share_i = \lfloor T / n \rfloor,\ r = T - n \cdot \lfloor T / n \rfloor,\ share_payer = \lfloor T / n \rfloor + r$$

For example, Rs 100 split among three people gives 34, 33 and 33, with the extra rupee assigned to the payer. Percentage splits use the floor of T × p / 100 for each person and assign the remainder in the same way. Exact splits must sum to the total, and line-item splits divide each item among the people who shared it and require the items to sum to the total. The calculation module is pure and is checked twice more: by the controller before saving and by a validation hook in the database model.

4.4 Settlement Algorithm (Min-Cash-Flow)

4.4.1 Net balance

For each member u of a group, the net balance is computed from all non-deleted expenses and all settlement payments already marked as paid:

$$net(u) = \Sigma\ paid(u) - \Sigma\ owed(u) + \Sigma\ settled_out(u) - \Sigma\ settled_in(u)$$

A positive net means the group owes the member (a creditor); a negative net means the member owes the group (a debtor). Because every rupee paid is owed by someone, the nets always sum to zero; the algorithm verifies this and refuses to run otherwise, so a defect elsewhere can never create or lose money.

4.4.2 Greedy matching

Figure 4.3 shows the algorithm as a flowchart, and the pseudocode is given below.

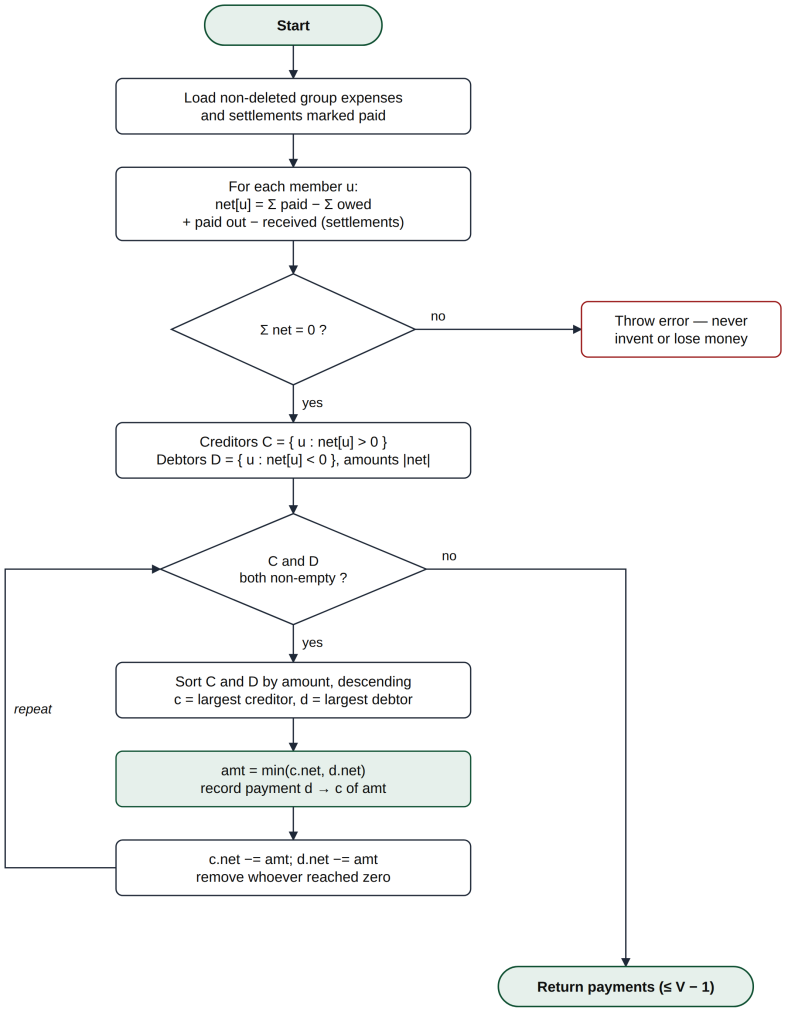


Figure 4.3: Flowchart of the Min-Cash-Flow settlement algorithm

```
MIN-CASH-FLOW(net)
  C ← { (u, net[u]) : net[u] > 0 }          // creditors
  D ← { (u, −net[u]) : net[u] < 0 }         // debtors
  if Σ C ≠ Σ D then error "balances do not net to zero"
  payments ← [ ]
  while C ≠ ∅ and D ≠ ∅ do
      sort C and D by amount, descending
      c ← C[0];  d ← D[0]
      amt ← min(c.amount, d.amount)
      append (d.user → c.user, amt) to payments
      c.amount ← c.amount − amt;  d.amount ← d.amount − amt
      remove c from C if c.amount = 0
      remove d from D if d.amount = 0
  return payments
```

4.4.3 Correctness and bound

Termination and bound. In every iteration, amt equals the smaller of the two amounts, so at least one of the two members reaches zero and leaves the pool. After k payments at least k members have left. Since the balances sum to zero, when only one member remains that member's balance must also be zero. Therefore the algorithm stops after at most V − 1 payments for V members with non-zero balances [1].

Correctness. Each payment reduces a debtor's debt and a creditor's credit by the same amount, so the sum of balances stays zero and every amount stays an integer. When the loop ends, both lists are empty and every balance is exactly zero.

Complexity. There are at most V − 1 iterations and each sorts at most V entries, giving $O(V^2 \log V)$ time and O(V) space. For groups of up to a few dozen members this is instantaneous; a binary heap would reduce it to O(V log V) for very large groups.

Optimality. The greedy result is not always the absolute minimum: when a subset of members already nets to zero among themselves, fewer payments may exist. Computing the true minimum is NP-hard [1], [2]; the V − 1 bound with polynomial running time is the accepted practical trade-off.

4.4.4 Worked example

The demonstration group "Murree Trip" has five members and seven expenses totalling Rs 49,600. Table 4.2 shows the net balances computed by the system and Table 4.3 shows each iteration of the algorithm.

Table 4.2: Net balances of the Murree Trip group

Member	Net balance (Rs)	Role
Sara Khan	+11,480	Creditor
Bilal Ahmed	+7,580	Creditor
Azeem Shahid	−3,520	Debtor
Hamza Tariq	−7,420	Debtor

Member	Net balance (Rs)	Role
Ali Raza	−8,120	Debtor
Total	**0**	

Table 4.3: Iterations of Min-Cash-Flow on the Murree Trip group

Step	Largest creditor	Largest debtor	Payment and remainder
1	Sara +11,480	Ali −8,120	Ali → Sara Rs 8,120; Sara +3,360, Ali settled
2	Bilal +7,580	Hamza −7,420	Hamza → Bilal Rs 7,420; Bilal +160, Hamza settled
3	Sara +3,360	Azeem −3,520	Azeem → Sara Rs 3,360; Sara settled, Azeem −160
4	Bilal +160	Azeem −160	Azeem → Bilal Rs 160; both settled

The group is settled in four payments, which equals the bound $V - 1 = 4$. Repaying each share of each bill individually would have required 20 separate transfers, so the optimiser saves 16 transfers.

4.5 Receipt Processing Pipeline

Figure 4.4 shows the sequence of interactions when a receipt is scanned. The upload request returns HTTP 202 (Accepted) immediately, and extraction runs in the background while the client polls for the result. The vision model is called with **structured output**: the response must match a fixed schema (store, date, items with name, quantity and line price, and total), so it cannot return free text or malformed JSON. The prompt fixes the domain rules: whole rupees, line totals rather than unit prices, subtotal and tax rows skipped, and null rather than a guess for unreadable values. If the call fails for any reason, Tesseract.js produces the same structure from raw text using line-pattern rules, and the receipt records which engine was used.

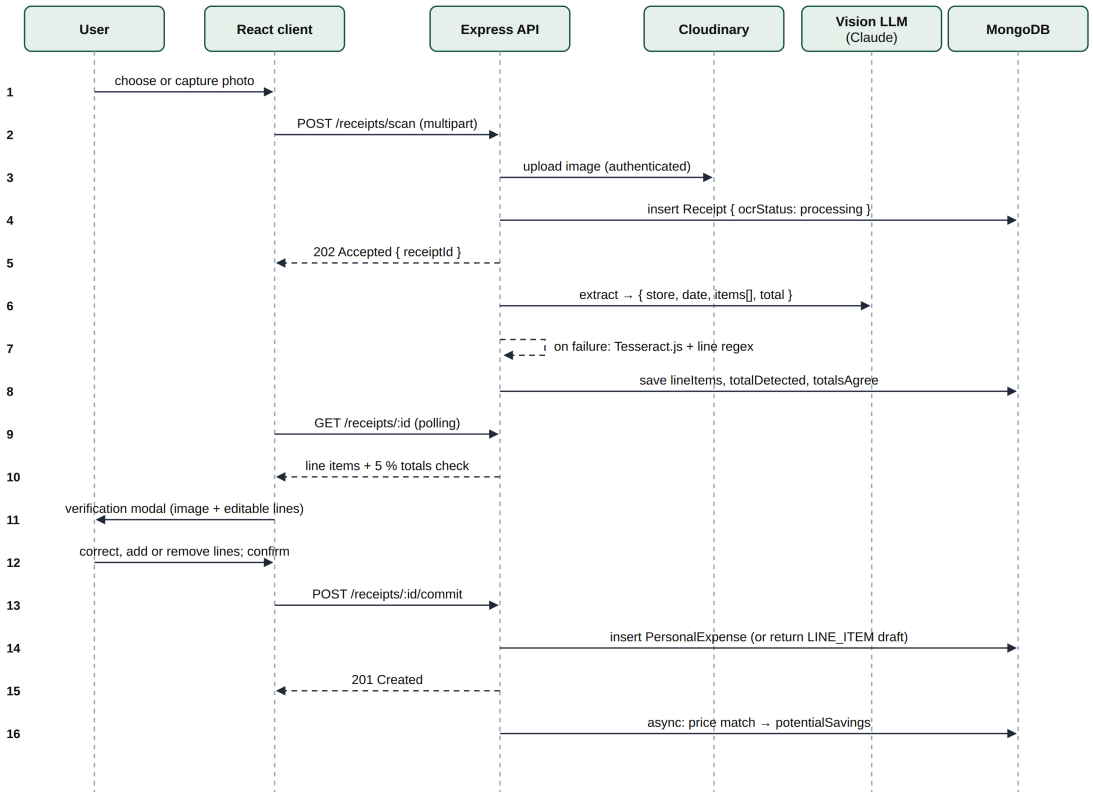


Figure 4.4: Sequence diagram of receipt scanning and verification

After extraction, the sum of the line prices is compared with the printed total; a difference above 5% produces a warning that names both values. Nothing is written as a transaction until the user confirms the lines in the verification dialog. Committing to a group does not create the expense; it returns a line-item draft so the user still chooses who shared each item.

4.6 Price Matching and Savings Suggestions

Each verified line item is matched against the benchmark collection in four steps:

1. **Normalisation** — the name is lower-cased and pack sizes, units, punctuation and filler words are removed, so "COOKING OIL 3LTR @" and "Cooking Oil 3 Litre" produce the same key.
2. **Alias lookup** — an indexed exact match against known receipt spellings, for example "anday" → Eggs (dozen) or "surf excel" → Detergent 1 kg.
3. **Fuzzy match** — otherwise the Dice coefficient [5] is computed, and a match is accepted only at 0.8 or above.
4. **Size check** — pack sizes are parsed separately (kg ↔ g, L ↔ ml) and must agree within 2%, so a 5 kg bag is never compared with a 1 kg bag.

$$Dice(a, b) = 2 \cdot |bigrams(a) \cap bigrams(b)| \,/\, (|bigrams(a)| + |bigrams(b)|)$$

Among matching products, the cheapest platform is selected; if the user paid more, the saving (multiplied by quantity) is recorded and up to three cheaper alternatives are attached to the transaction. Matching runs after the save has been acknowledged, so it never delays the user.

4.7 Multi-Currency Conversion

Trips abroad produce bills in other currencies. The user may enter a group expense in one of 15 foreign currencies (for example AED, SAR, USD or GBP) with up to two decimal places. The server converts every amount to rupees at the current rate R (rupees per unit of the foreign currency) and stores the expense in whole rupees, so balances and settlement are unchanged. Rates come from a free, keyless exchange-rate service [22], refreshed every day at 03:00 and stored in the database as a fallback. Each converted value is rounded separately, so their sum may differ from the converted total by a rupee or two; that difference is given to the largest payer (or the largest row of an exact or line-item split):

$$T_PKR = round(T \times R),\ p_i = round(a_i \times R),\ p_largest \mathrel{+}= T_PKR - \Sigma\ p_i$$

Table 4.4 shows the conversion of a desert safari on the demonstration Dubai trip at the rate on 30 October 2026. The original amounts and the rate are kept in an `fx` sub-document, so the expense list can show "AED 412.5" under the rupee amount, and editing the expense later keeps the rate it was added at.

Table 4.4: Conversion of AED 412.50 shared equally by four members at 1 AED = Rs 75.2955

Value	In AED	In rupees	Calculation
Total, paid by Azeem	412.50	31,059	412.50 × 75.2955 = 31,059.39, rounded
Share of Sara, Bilal and Ali	103.125 each	7,764 each	⌊31,059 / 4⌋
Share of Azeem (payer)	103.125	7,767	7,764 + remainder 3 (BR-03)
Sum of shares	412.50	31,059	Equals the stored total (BR-02)

4.8 Savings Goal Progress

A savings goal stores only its target and the list of deposits and withdrawals; everything the user sees is computed when the goal is read, in the same spirit as group balances. For a goal of target A due on date D, created on date C, with saved amount S on date t:

$$percent = \lfloor 100 \cdot S / A \rfloor,\ monthly = \lceil (A - S) / \lceil days_left / 30 \rceil \rceil$$

$$on\ track \Leftrightarrow S \geq A \cdot (t - C) / (D - C)$$

The "on track" test compares the saved amount with a straight line from nothing on the day the goal was created to the full target on its date. When the saved amount first reaches the target, the goal records the date and the owner receives a single GOAL_REACHED

notification; a withdrawal that would take the goal below zero is refused.

4.9 Security Design

Authentication uses two tokens with different lifetimes and storage (Table 4.5). Figure 4.5 shows sign-in, transparent token refresh and reuse detection.

Table 4.5: Token design

Token	Lifetime	Storage	Purpose
Access token (JWT)	15 minutes	Browser memory only (never localStorage)	Sent as a Bearer header with every API request
Refresh token	7 days	HttpOnly, SameSite=Strict, Secure cookie scoped to /api/v1/auth; only its hash stored server-side	Obtains a new access token; rotated on every use

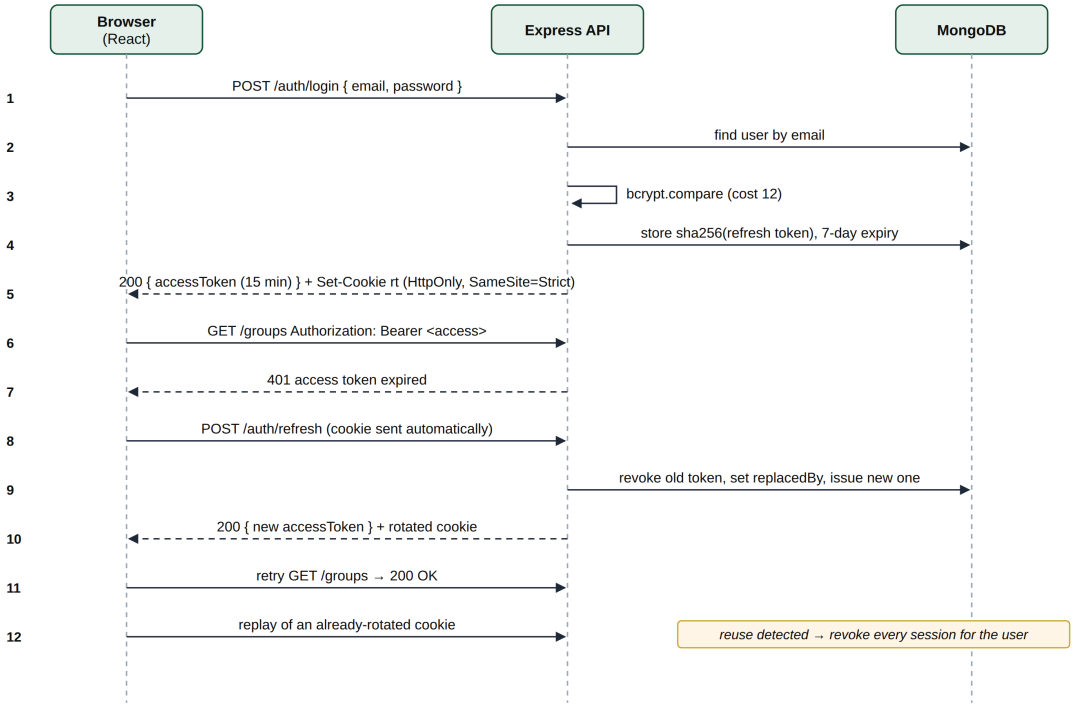


Figure 4.5: Sequence diagram of sign-in, token refresh and reuse detection

If a refresh token that has already been rotated is presented again, it can only be a copy, so all of that user's sessions are revoked. Further controls include bcrypt hashing at cost 12 [8]; identical responses for unknown emails and wrong passwords; single-use password reset tokens stored as SHA-256 hashes with a 30-minute expiry; Zod validation of every request body, parameter and query; per-route and global rate limits; Helmet security headers; CORS restricted to the client origin; private, signed receipt image URLs; and a start-up guard that refuses to run in production with missing, weak or placeholder secrets. Super administrators

are defined by the `ADMIN_EMAILS` server setting rather than a database field, so the role cannot be granted through the API. These controls address the OWASP categories of broken access control, cryptographic failures, injection and authentication failures [9].

4.10 API Design

The API exposes 79 REST endpoints under the prefix `/api/v1`, grouped by resource (authentication, groups and invitations, expenses, settlements, personal transactions and budgets, savings goals, receipts, reports and notifications, categories, users, exchange rates, the price book and administration). Every response uses one of two envelopes:

```
{ "success": true,  "data": { ... } }
{ "success": false, "error": { "code": "SPLIT_INVALID", "message": "You're Rs 50
short" } }
```

Machine-readable error codes (for example `NOT_A_MEMBER`, `TOKEN_REUSED`, `SETTLEMENT_IN_PROGRESS`) let the client react precisely, while the human-readable message is shown to the user unchanged. The complete endpoint list is given in Appendix A.

4.11 User Interface Design

The interface uses a clean, card-based layout with a fixed top navigation (Dashboard, Groups, Personal, Goals, Scan and Reports, plus Admin for super administrators) and a notification bell. Colour carries meaning consistently: green for money owed to the user or within budget, amber for budgets above 80%, and red for debts and overspending. Every list implements loading, empty and error states; destructive actions ask for confirmation; and amounts are always formatted as "Rs 4,200". Layouts collapse to a single column on narrow screens, and wide tables scroll horizontally within their cards.

CHAPTER 5
IMPLEMENTATION

This chapter describes the tools, project structure and implementation of the main modules, and presents the user interface.

5.1 Technology Stack

Table 5.1: Technologies used

Layer	Technology	Role in the system
Frontend	React 19, Vite	Component-based single-page application with fast development builds
Frontend	React Router	Client-side routing across 20 routes with protected routes
Frontend	Tailwind CSS	Utility-first, responsive styling
Frontend	Recharts, Lucide	Charts on the Reports page; icon set
Frontend	Axios	HTTP client with interceptors for tokens and silent refresh
Backend	Node.js, Express 5	REST API server and middleware pipeline
Database	MongoDB, Mongoose	Document storage, aggregation pipelines, schemas and validation
Validation	Zod	Schema validation of every request
Security	jsonwebtoken, bcryptjs, Helmet, express-rate-limit	Tokens, password hashing, headers and rate limiting
Identity	Google Identity Services / google-auth-library	Google OAuth 2.0 sign-in
AI / OCR	Anthropic Claude vision (SDK), any OpenAI-compatible vision model, Tesseract.js	Structured receipt extraction, a free AI option, and an offline fallback
Files	Multer, Cloudinary	Upload handling and private image storage
Jobs	node-cron	Nightly recurring-transaction job (02:00) and exchange-rate refresh (03:00)
Exchange rates	ExchangeRate-API (open access)	Daily rupee rates for 15 currencies, no API key
Exports	json2csv, ExcelJS, PDFKit	CSV, XLSX and PDF generation
Email	Nodemailer, Brevo SMTP	Password reset and invitation emails
Testing	Vitest, Supertest, mongodb-memory-server	Unit, API and property-based tests on an in-memory database
Hosting	Render, MongoDB Atlas	One web service for the API and the built client; managed database

Layer	Technology	Role in the system
CI/CD	GitHub Actions	Runs every test and the client build on each push; Render deploys only if they pass

5.2 Project Structure

The code is organised as a monorepo with separate client and server packages. The server contains approximately 5,800 lines of application code and 3,900 lines of tests; the client contains approximately 6,000 lines across 19 pages and 26 reusable components.

```
smartsplit/
├── client/                 React + Vite application
│   └── src/
│       ├── pages/          one component per route (19)
│       ├── components/     reusable UI (26)
│       ├── api/            Axios instance and API functions
│       └── context/        AuthContext, CategoriesContext
└── server/                 Express API
    └── src/
        ├── models/         Mongoose schemas (14)
        ├── routes/         routers + Zod schemas per resource
        ├── controllers/    request handlers
        ├── services/       settlement, splitCalculator, balances, ocr,
        │                   priceMatch, currency, fx, recurring, reports, ...
        ├── middleware/     auth, membership, validate, rate limits, errors
        └── tests/          Vitest + Supertest suites (21 files)
```

5.3 Backend Implementation

5.3.1 Settlement service

The settlement service implements the algorithm of Section 4.4 as a pure function. Its core loop is shown below.

```
while (creditors.length && debtors.length) {
  creditors.sort(byLargest);
  debtors.sort(byLargest);

  const creditor = creditors[0];
  const debtor = debtors[0];
  const amount = Math.min(creditor.net, debtor.net);

  payments.push({ from: debtor.userId, to: creditor.userId, amount });

  creditor.net -= amount;
  debtor.net -= amount;

  // At least one of the two is now square and leaves the pool.
  if (creditor.net === 0) creditors.shift();
  if (debtor.net === 0) debtors.shift();
}
```

The same net balances are also computed inside MongoDB with an aggregation pipeline ($unwind and $group over the paidBy and splits arrays, combined with a $facet), which is used by the live routes for efficiency [12]. An automated test runs both implementations on the same ledger and asserts identical results, preventing the two from drifting apart.

5.3.2 Split calculator

The split calculator distributes an integer total with the floor-and-remainder rule:

```
function distribute(total, userIds, preferredUserId) {
  const headCount = userIds.length;
  const base = Math.floor(total / headCount);
  const shares = new Map(userIds.map((id) => [String(id), base]));

  const leftover = total - base * headCount;
  if (leftover > 0) {
    const target = preferredUserId && shares.has(String(preferredUserId))
      ? String(preferredUserId) : String(userIds[0]);
    shares.set(target, shares.get(target) + leftover);
  }
  return shares;
}
```

5.3.3 Authentication and authorisation

The `requireAuth` middleware verifies the Bearer access token and loads the user. The `requireMember` middleware loads the group from the URL identifier and looks up the caller's own active membership using the authenticated user identifier only; a non-member receives HTTP 403 and an unknown group HTTP 404. `requireAdmin` additionally checks the admin role; it guards member management and, because recording a settlement closes the group, settlement recording as well. Refresh-token rotation with reuse detection is implemented as follows:

```
const existing = await RefreshToken.findOne({ tokenHash: hash(token) });
if (!existing) return { error: 'INVALID' };

// A rotated token presented again is evidence the cookie leaked.
if (existing.revokedAt && existing.replacedBy) {
  await RefreshToken.updateMany({ userId: existing.userId, revokedAt: null },
                                { revokedAt: new Date() });
  return { error: 'REUSED' };
}
if (existing.revokedAt || existing.expiresAt < new Date()) return { error:
'INVALID' };

const next = await issueRefreshToken(existing.userId);
existing.revokedAt = new Date();
existing.replacedBy = hash(next.token);
await existing.save();
```

5.3.4 Receipt OCR and price matching

Uploads are received in memory by Multer, limited to 8 MB and image types, and stored on Cloudinary as authenticated assets. The OCR service tries up to three engines in order: the Claude vision model through the Anthropic SDK with a Zod-defined output format; otherwise any OpenAI-compatible vision model (for example a free model on OpenRouter or Google AI Studio), whose reply is parsed leniently and validated with the same schema; and finally Tesseract.js with its English language data bundled locally, so it needs no download and a failure can never stop the server. For the offline engine, a regular expression extracts lines such as "Cooking Oil 3L 1,745", "2 x Milk 220" and quantity–rate–amount columns. The price-matching service implements normalisation, alias lookup, the Dice coefficient (shown below) and size parsing.

```
export function similarity(leftText, rightText) {
  if (leftText === rightText) return 1;
  if (leftText.length < 2 || rightText.length < 2) return 0;
  const first = bigrams(leftText);
  const second = bigrams(rightText);
  let shared = 0;
  for (const [pair, count] of first) shared += Math.min(count, second.get(pair) ||
0);
  return (2 * shared) / (leftText.length - 1 + (rightText.length - 1));
}
```

5.3.5 Multi-currency expenses

The currency service is a pure function that receives a validated expense in its entry currency and the rate, and returns the same expense in rupees together with the original figures. Payers, exact shares and line items are first checked in cents of the entry currency, so a message such as "Payers are AED 40 short" is given in the currency the user typed. The helper below rounds each value and moves the rounding difference to one chosen row:

```
function toRupees(values, rate, totalRupees, target) {
  const rupees = values.map((value) => Math.round(value * rate));
  rupees[target] += totalRupees - rupees.reduce((sum, value) => sum + value, 0);
  return rupees;
}
```

The exchange-rate service keeps the latest rates in memory for 12 hours and in the `fx_rates` collection. If the provider cannot be reached, the saved copy is used and marked as stale, and a failed attempt is not repeated for ten minutes; only when no rates were ever saved does the API answer HTTP 503 with the code `FX_UNAVAILABLE`.

5.3.6 Savings goals

The goals controller exposes six endpoints (list, create, edit, delete, add a contribution and remove a contribution). Every query includes the owner's identifier, so a goal belonging to someone else answers HTTP 404. Progress is computed by a pure `progressOf(goal, now)`

function that implements the formulas of Section 4.8, and is unit-tested with fixed dates.

5.3.7 Recurring transactions, reports and notifications

A node-cron job runs every night at 02:00 and calls `runDueRecurring(now)`, which finds every template whose next run date has passed, writes one entry per missed period (so a server that was offline for three months catches up with three entries) and advances the schedule; a 120-iteration guard prevents runaway loops. Reports are computed with aggregation pipelines and exported with json2csv, ExcelJS and PDFKit. Notifications are written as one row per recipient on four events; the notification function catches its own errors so that a failure can never cancel the money operation that triggered it.

5.4 Frontend Implementation

The client is built from function components and hooks. An `AuthContext` holds the signed-in user and keeps the access token in memory; on start-up it attempts a refresh so a page reload keeps the session. A shared Axios instance attaches the token, unwraps the response envelope and, on HTTP 401, performs a single refresh shared by all concurrent requests before retrying them. Pages never call Axios directly; they use typed functions in the `api` folder. A `useAsync` hook and a shared `States` component provide loading, empty and error states for every list, and the Reports page is lazy-loaded to keep the initial bundle small. An error boundary around each page reloads the application once if a newer deployment has replaced the page's code, instead of showing a blank screen.

5.5 User Interface

Figures 5.1 to 5.16 show the main screens of the application running with the demonstration data.

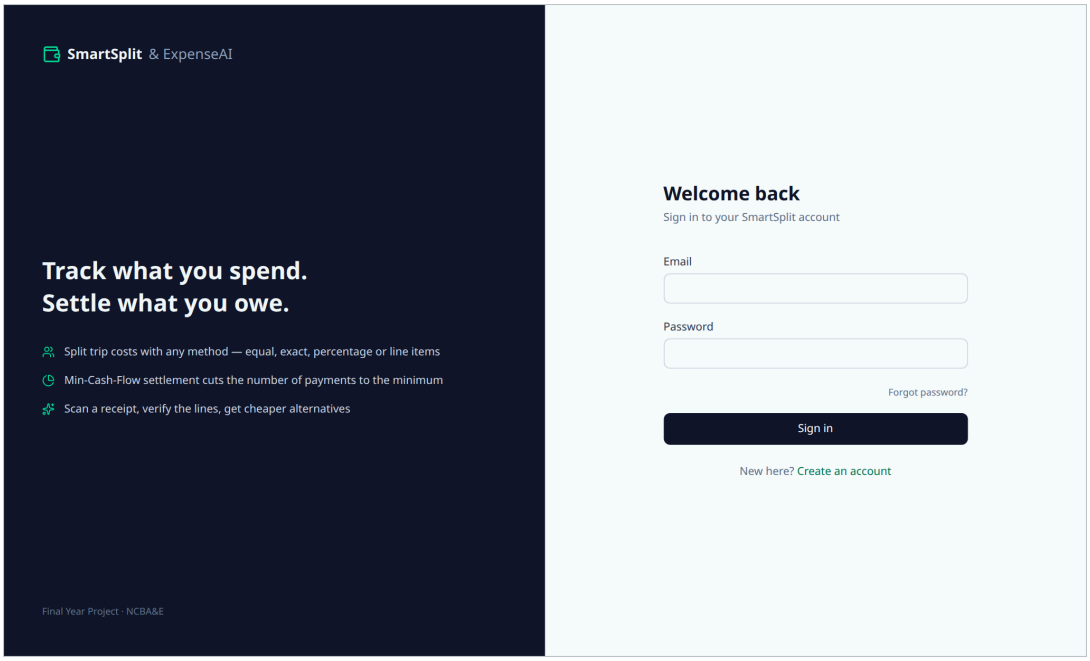


Figure 5.1: Sign-in page

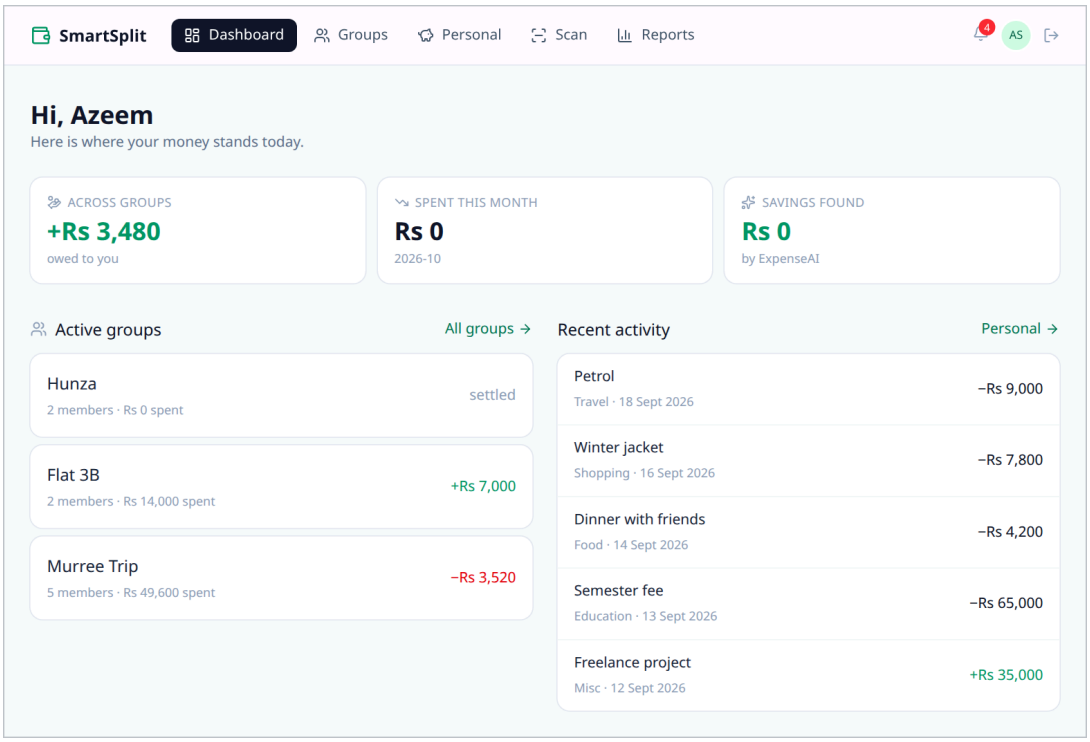


Figure 5.2: Dashboard combining both workspaces

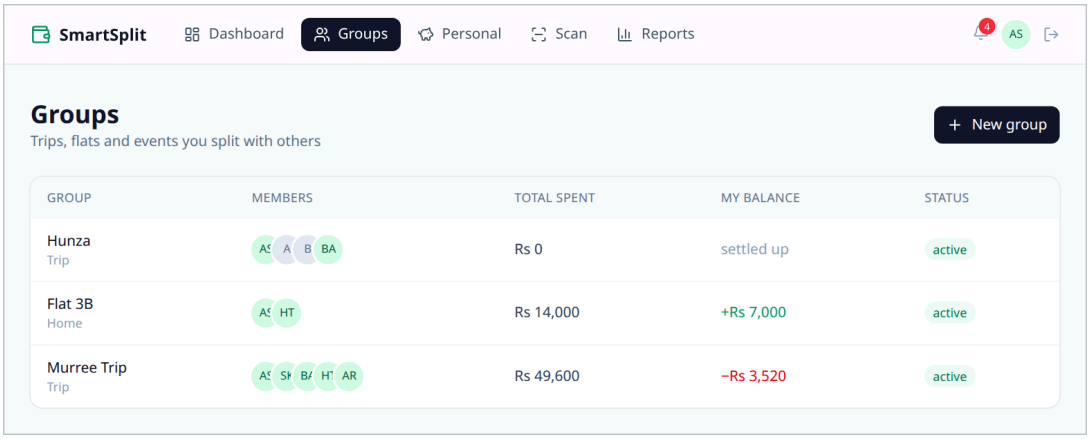


Figure 5.3: Groups list with total spent and personal balance

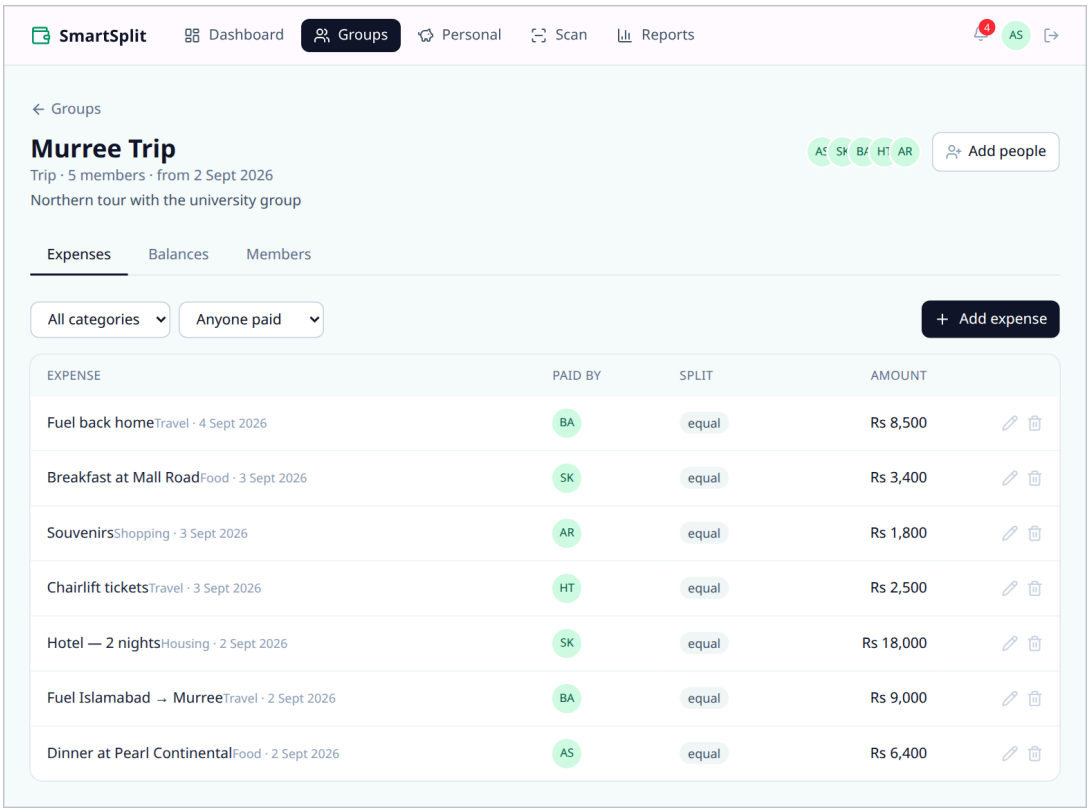


Figure 5.4: Group expenses of the Murree Trip

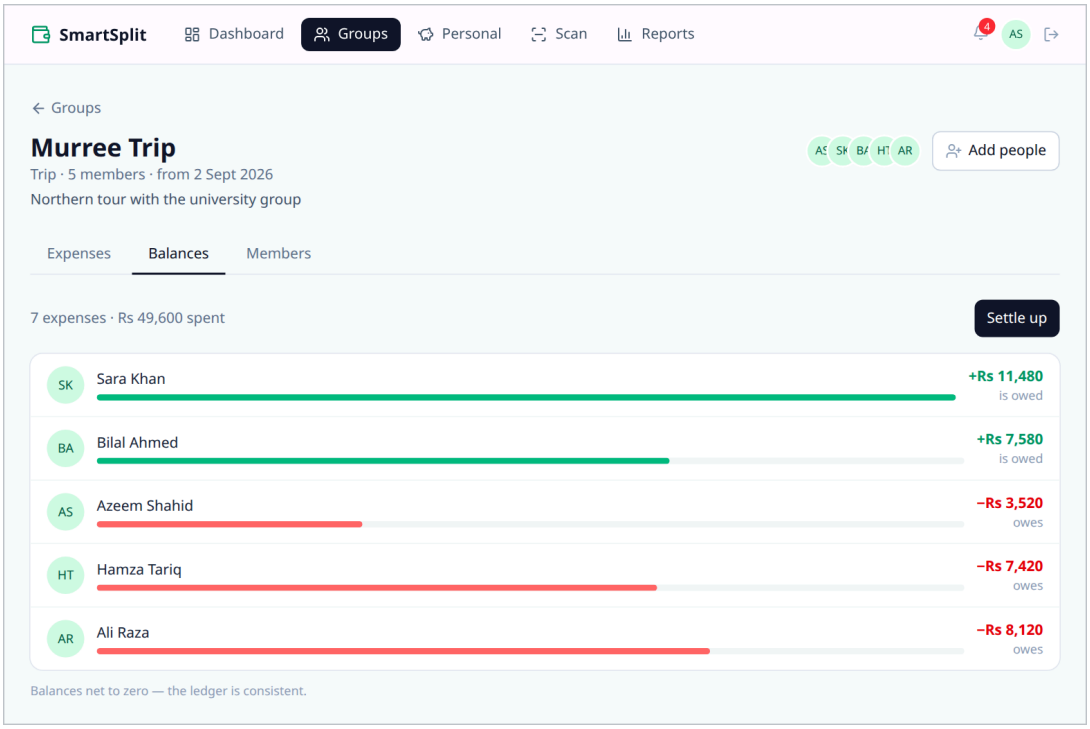


Figure 5.5: Group balances, which always net to zero

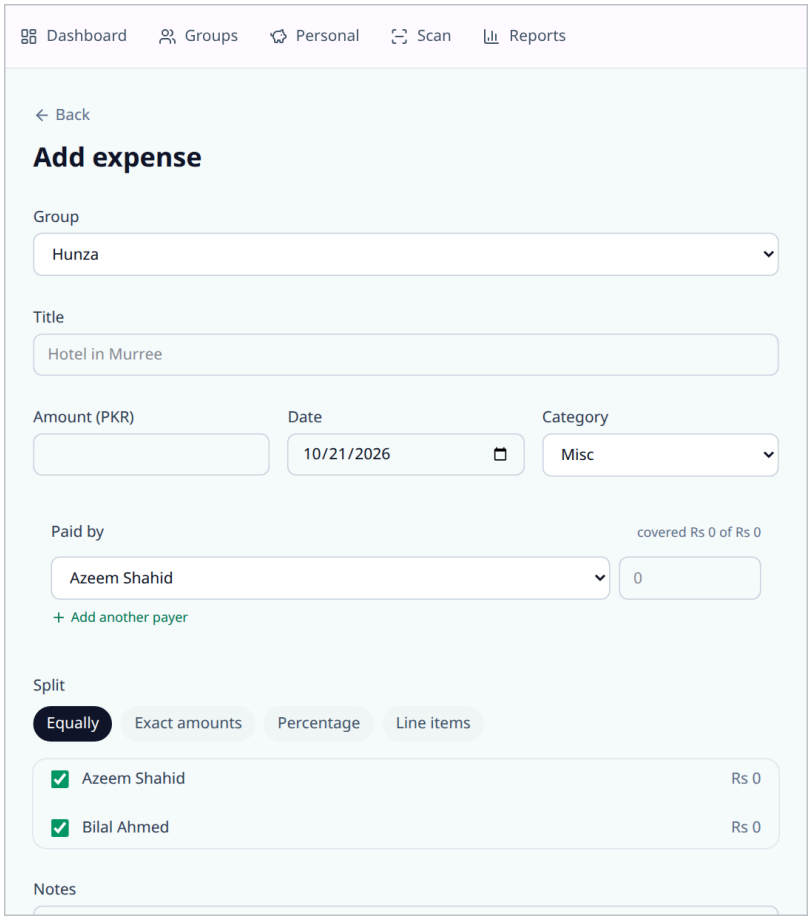


Figure 5.6: Add Expense form with multi-payer rows and split methods

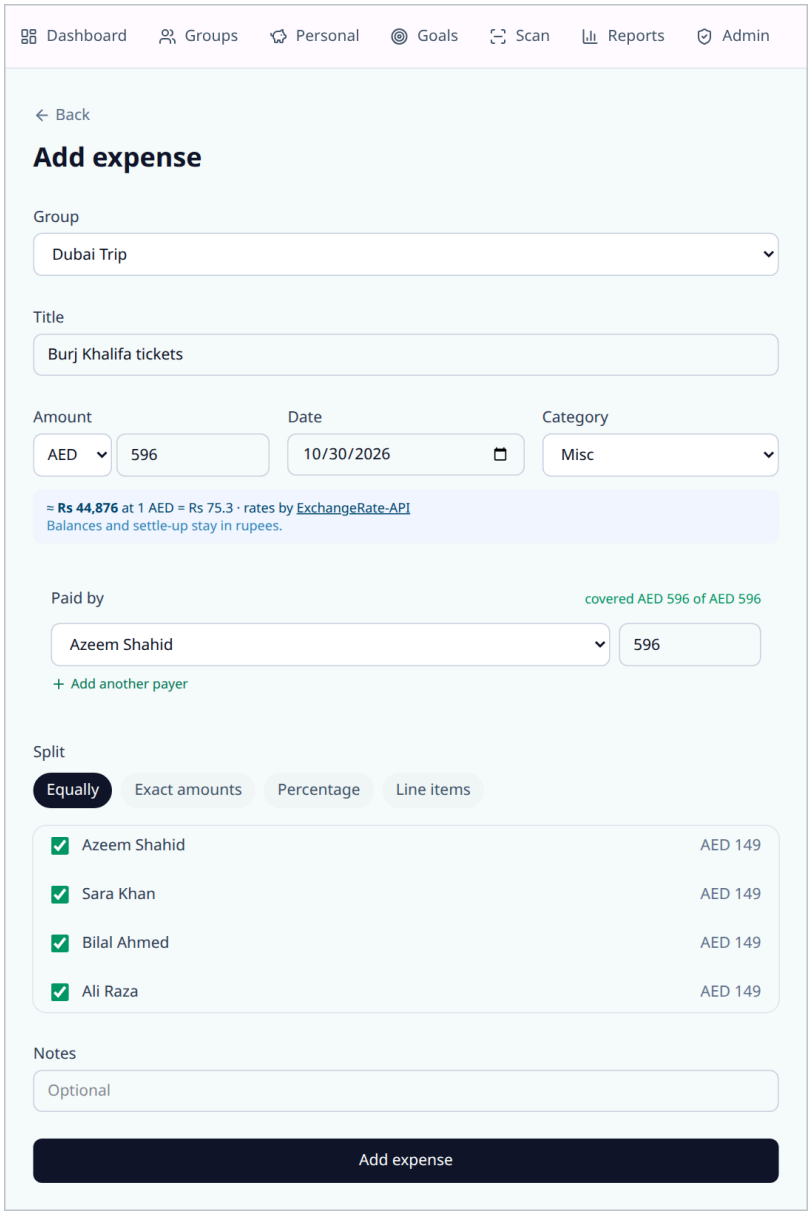


Figure 5.7: Expense entered in UAE dirhams with the live rupee preview

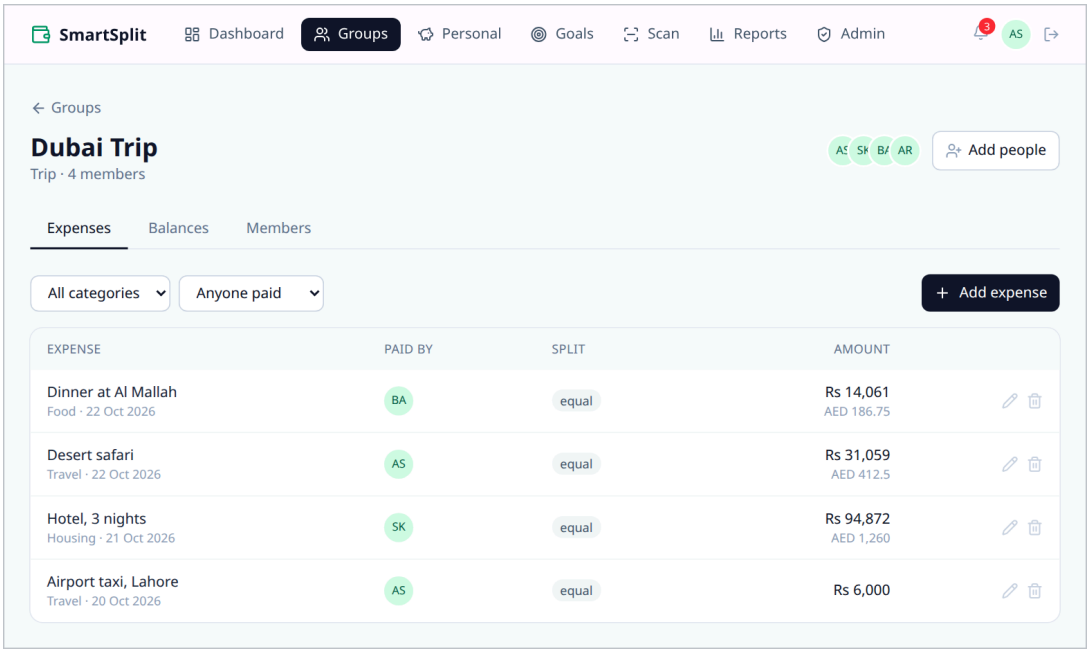


Figure 5.8: Dubai Trip expenses: rupee amounts with the original AED figures

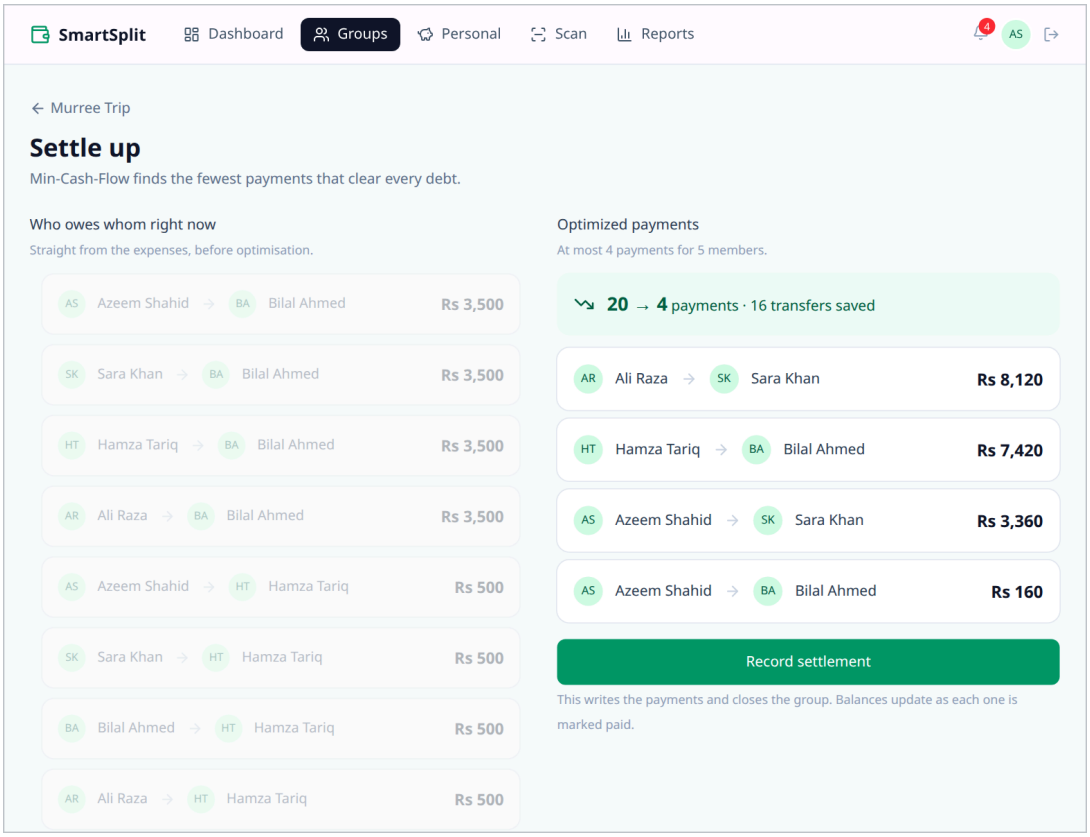


Figure 5.9: Settle Up after optimisation: 20 naive debts reduced to 4 payments

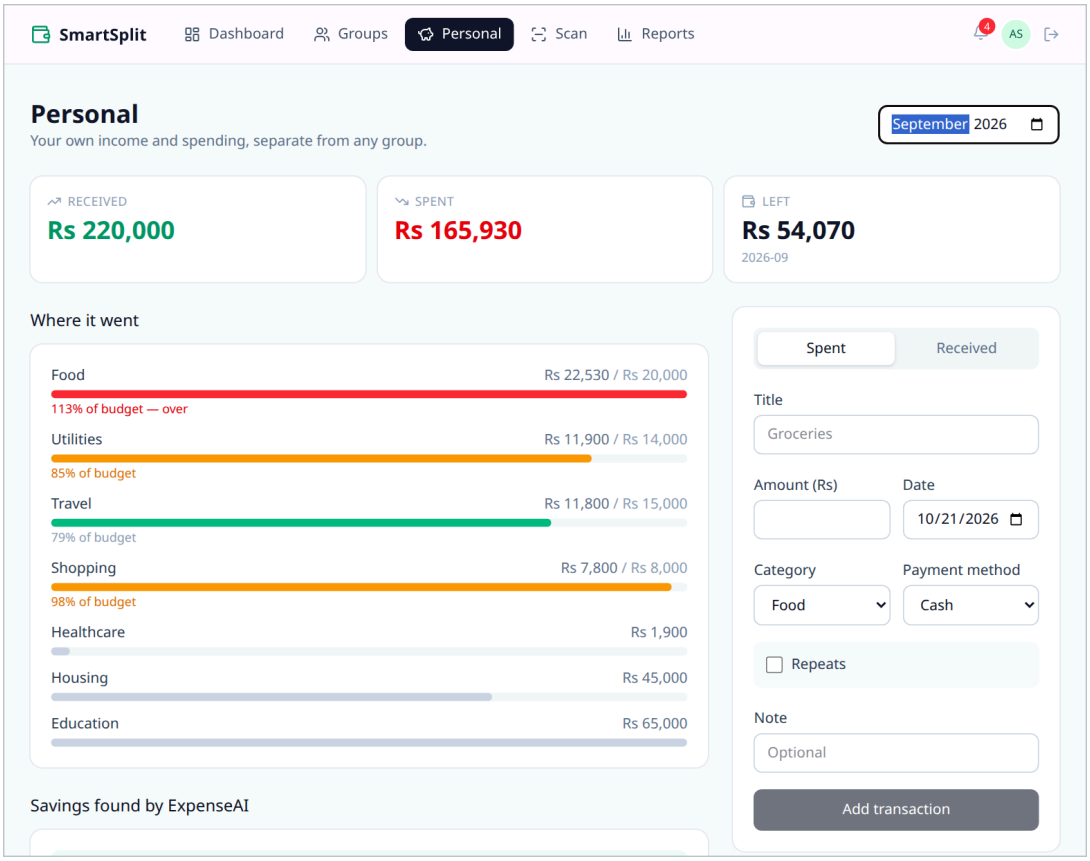


Figure 5.10: Personal workspace with budget bands and transaction form

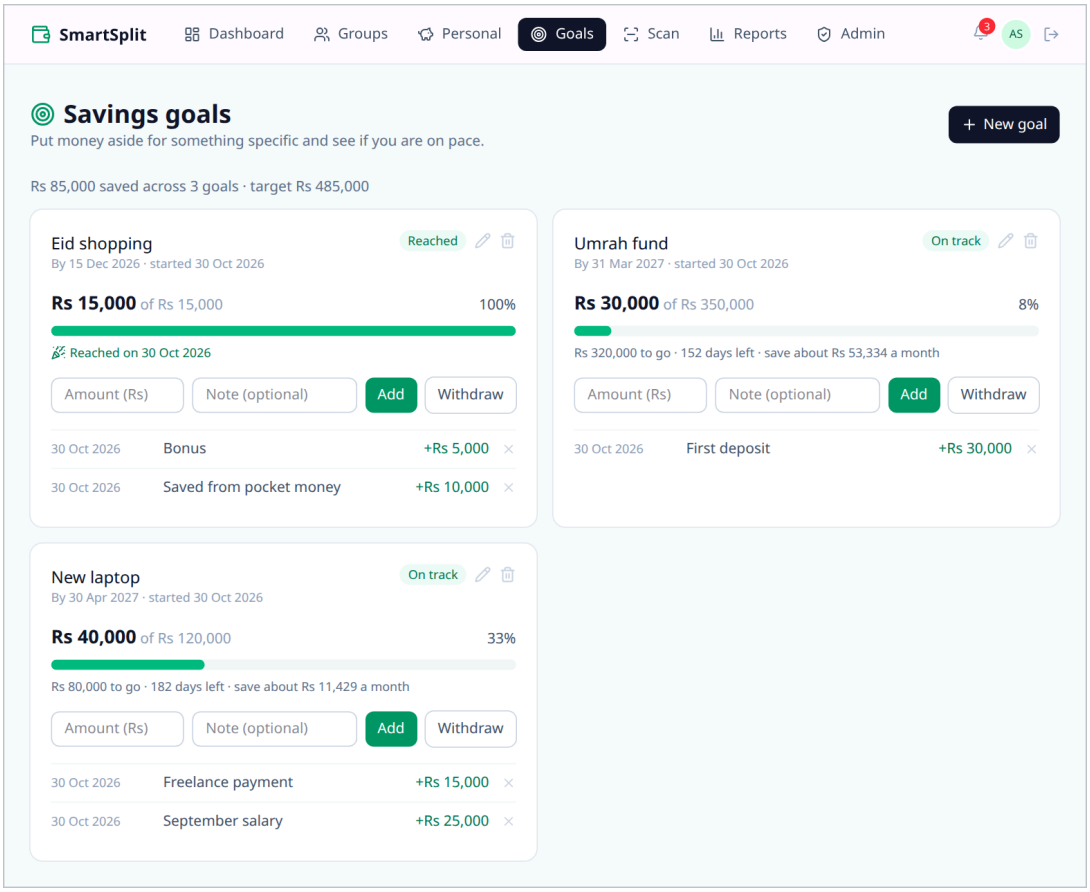


Figure 5.11: Savings goals with progress, pace and the monthly amount needed

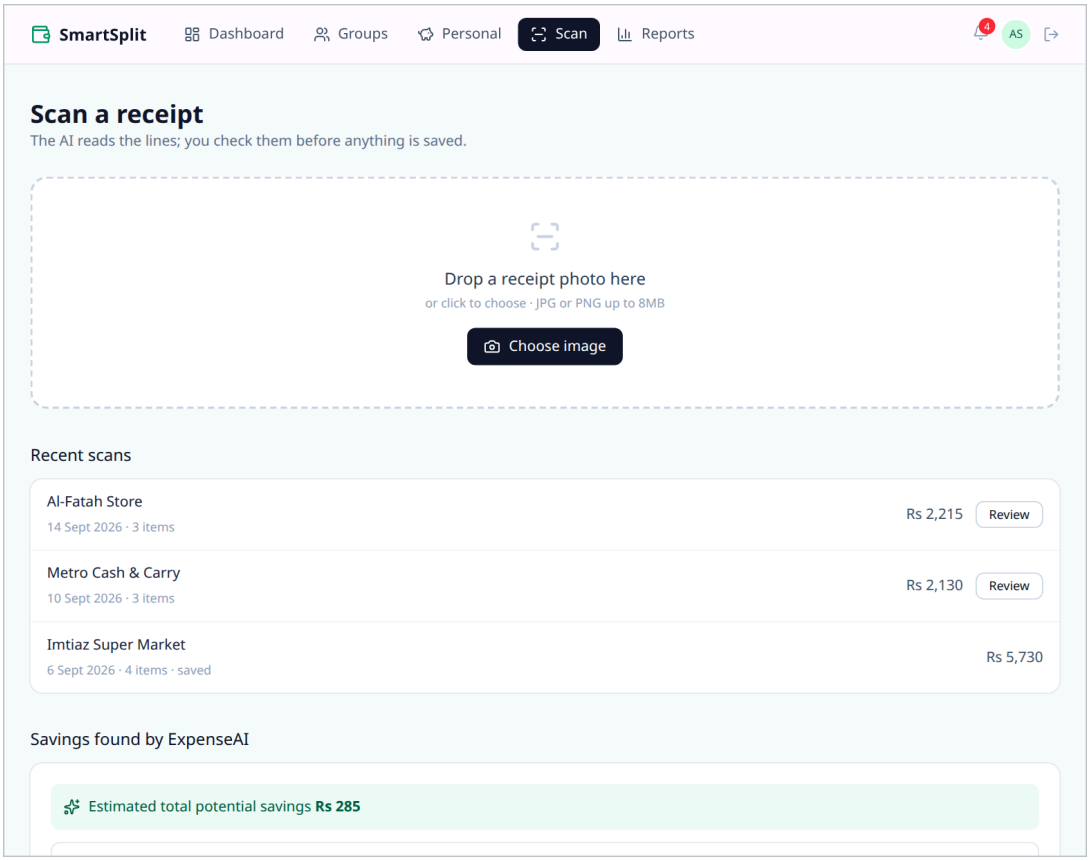


Figure 5.12: Receipt scanning page with recent scans and savings found

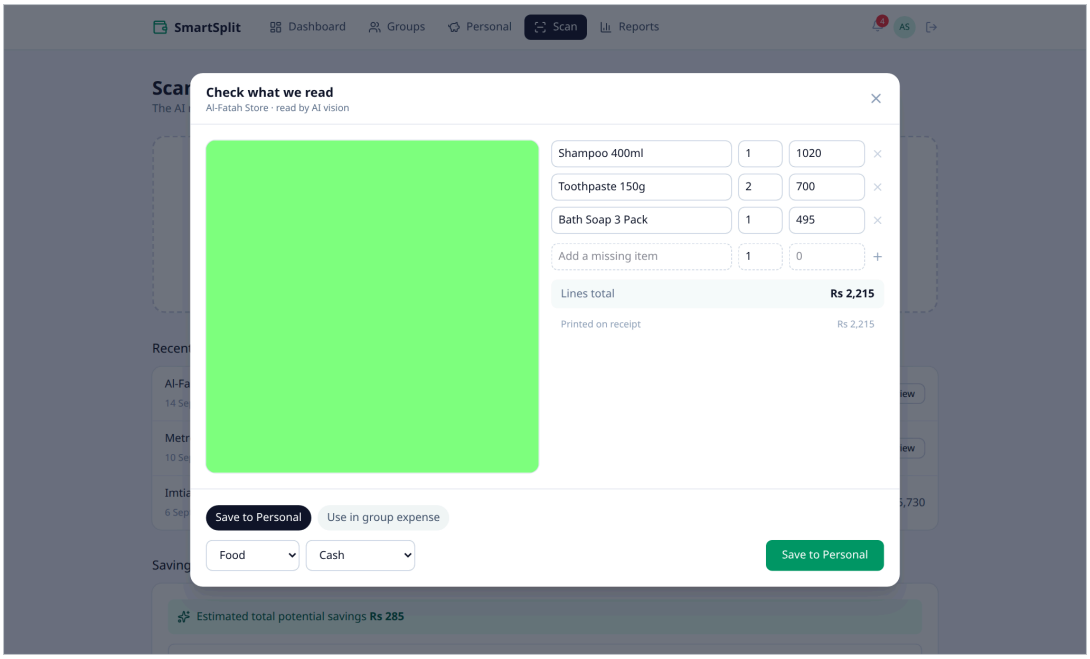


Figure 5.13: Human-in-the-loop verification dialog (demonstration data uses a placeholder image)

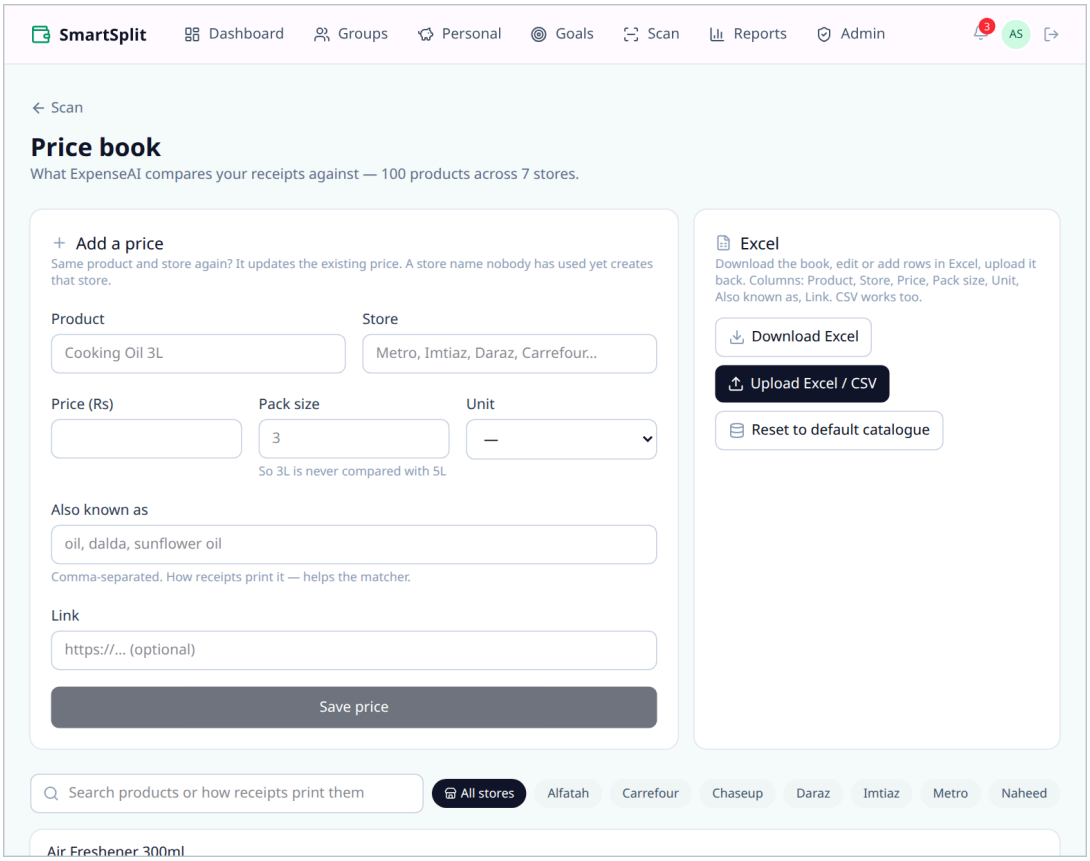


Figure 5.14: Price book of benchmark prices, maintained by super administrators

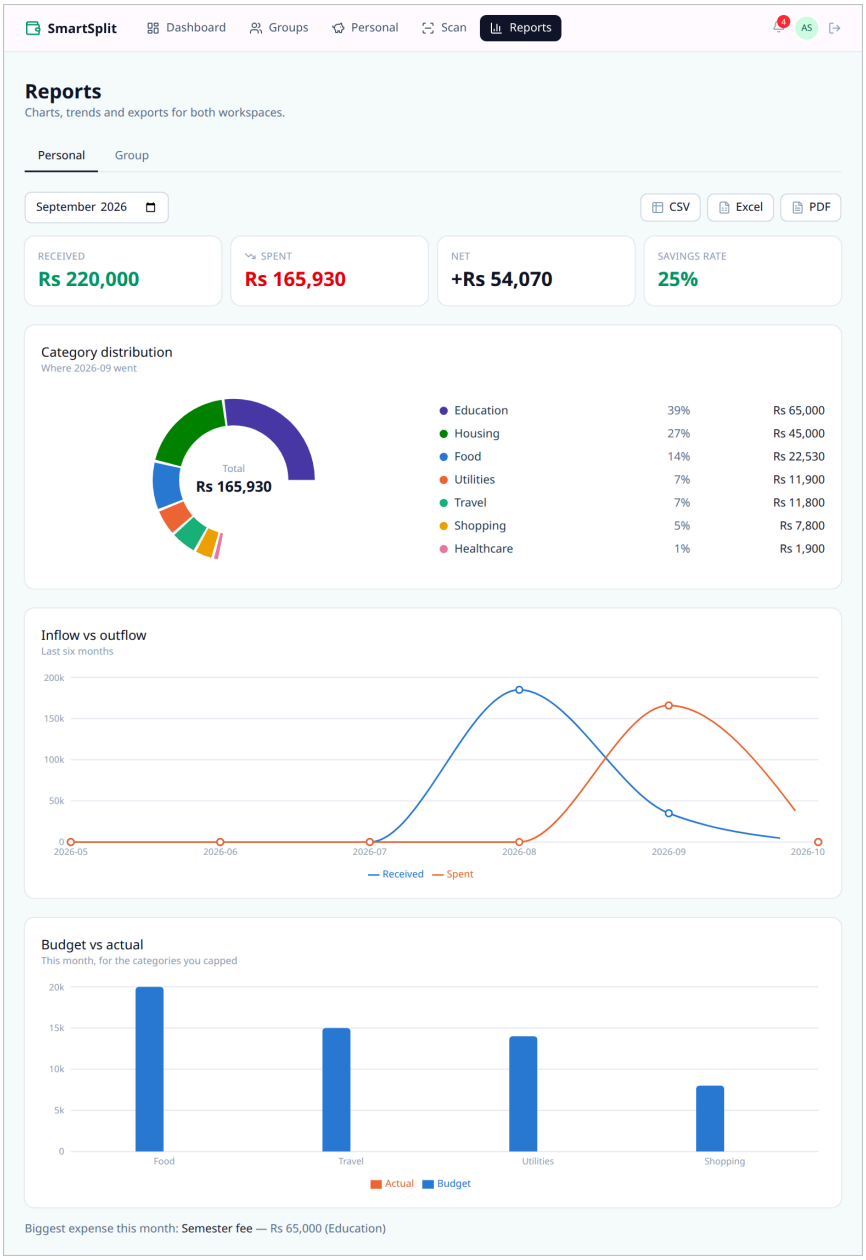


Figure 5.15: Personal reports: category distribution, inflow vs outflow, budget vs actual

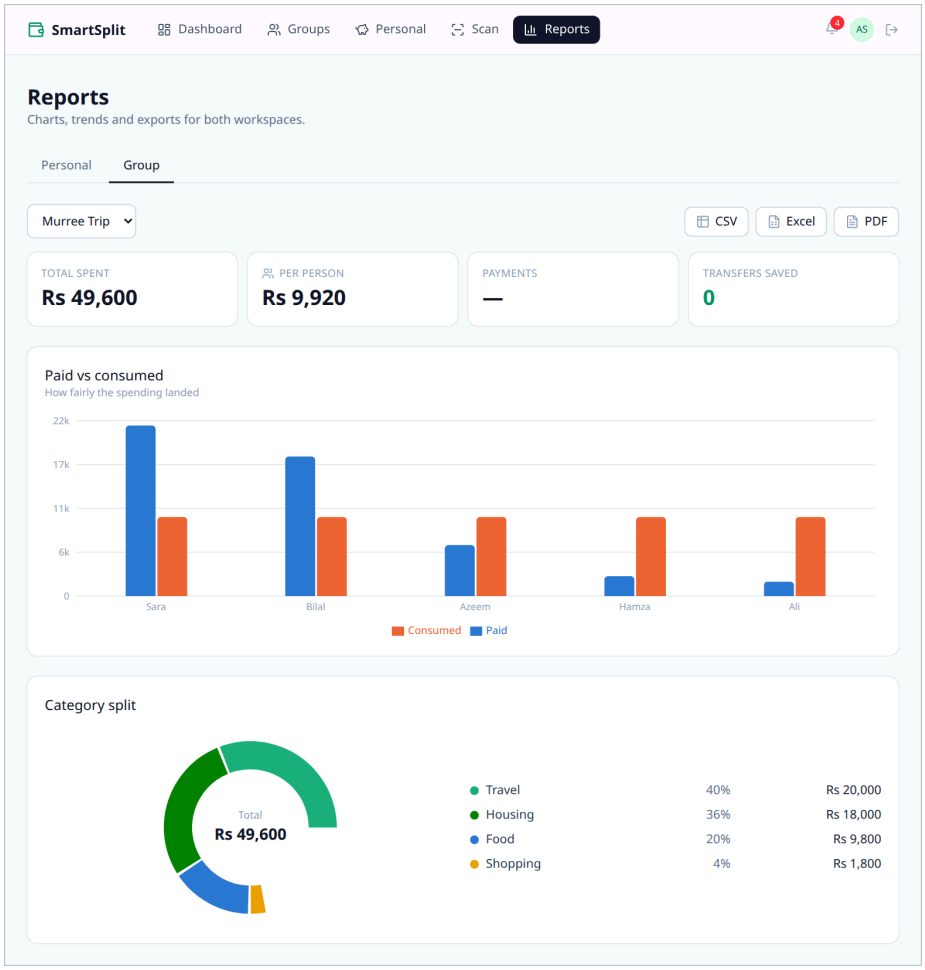


Figure 5.16: Group report: paid vs consumed per member and category split

5.6 Deployment

The system is live at https://smarsplit.onrender.com. It runs as a single Render web service: the build step installs both packages and builds the React client, and Express serves the built files together with the API. Serving both from one origin keeps the SameSite=Strict refresh cookie first-party and means the browser never makes a cross-origin request; any path outside `/api` returns the client, so deep links such as `/groups/:id` load correctly. The database is a MongoDB Atlas cluster, the health check is `/api/v1/health`, and all secrets are supplied as environment variables.

Deployment is automated and gated by tests. Every push to the main branch on GitHub starts a GitHub Actions workflow that runs the full server test suite and builds the client; Render deploys the new version only when that workflow passes (`autoDeployTrigger: checksPass`), so a failing change never reaches users. On the free tier the service pings its own health route every ten minutes to avoid sleeping during use, and on start-up it loads the default price catalogue if the price book is empty. A seed script (`npm run seed`) creates five demonstration users, two groups including the Murree Trip, twelve personal transactions, budgets, three scanned receipts and the price benchmark.

CHAPTER 6
TESTING AND EVALUATION

This chapter describes the testing strategy, summarises the automated test suite, presents selected test cases and the results of security review and defect tracking, and evaluates the settlement engine.

6.1 Testing Strategy

- **Unit testing** — pure functions (split calculator, settlement, price matching, environment guard) are tested in isolation with Vitest.
- **API (integration) testing** — Supertest sends real HTTP requests to the Express application connected to an in-memory MongoDB instance (mongodb-memory-server), verifying status codes, response envelopes, authorisation and database effects.
- **Property-based testing** [18] — the settlement engine is tested on hundreds of randomly generated groups to confirm that its guarantees hold for all inputs, not only hand-picked examples. A fixed random seed makes every failure reproducible.
- **Manual and exploratory testing** — every screen was exercised in the browser with the demonstration data, including narrow screen widths.

6.2 Test Suite Summary

The suite contains 288 tests in 21 files. All 288 tests passed on 30 October 2026 (Table 6.1), and the same suite runs automatically before every deployment.

Table 6.1: Automated test results by file

Test file	Area covered	Tests	Passed
auth.test.js	Register, login, refresh rotation, reuse detection, reset	18	18
profile.test.js	Profile update and password change	10	10
env.test.js	Production environment guard	7	7
health.test.js	Health route and 404 envelope	2	2
admin.test.js	Super admins, overview, default catalogue	4	4
groups.test.js	Groups, invites, inbox, member roles	23	23
membership.test.js	Direct add, invite, request-and-approve flows	26	26
splitCalculator.test.js	Four split methods and rounding rules	22	22
expenses.test.js	Group expense API, filters, balances	16	16
currency.test.js	Foreign-currency conversion and rounding	8	8
fx.test.js	Exchange rates, fallback, foreign expenses	12	12

Test file	Area covered	Tests	Passed
settlement.test.js	Min-Cash-Flow algorithm and property tests	12	12
settlementApi.test.js	Preview, record (admin), mark paid, consistency	11	11
personal.test.js	Transactions, budgets, recurring engine, summary	16	16
goals.test.js	Savings goals, contributions, progress	11	11
categories.test.js	Built-in and custom categories	19	19
receipts.test.js	Scan pipeline, verification, commit	18	18
ocr.test.js	OCR engines and offline receipt parser	7	7
priceMatch.test.js	Normalisation, similarity, size awareness	18	18
prices.test.js	Price book editing, Excel/CSV import, export	9	9
reports.test.js	Reports, CSV/XLSX/PDF exports, notifications	19	19
Total		**288**	**288**

6.3 Selected Test Cases

Table 6.2: Selected test cases

ID	Test case	Expected result	Result
TC-01	Split Rs 100 equally among three, payer included	Shares 34 / 33 / 33, payer receives 34	Pass
TC-02	Exact split totalling Rs 50 less than the bill	400 with "You're Rs 50 short"	Pass
TC-03	Percentages totalling 90%	400 with "Percentages add up to 90%, not 100%"	Pass
TC-04	Fractional amount (Rs 10.5)	400 validation error, nothing stored	Pass
TC-05	Non-member requests group expenses	403 NOT_A_MEMBER	Pass
TC-06	Plain member attempts to remove a member	403 NOT_ADMIN	Pass
TC-07	Replay of an already-rotated refresh token	401 TOKEN_REUSED and all sessions revoked	Pass
TC-08	Forgot password for unknown vs known email	Byte-identical responses	Pass
TC-09	Settle five-member reference group	Exactly 4 payments, all balances zero	Pass
TC-10	300 random groups of 2–10 members	Balances zero after payments; count ≤ V − 1	Pass
TC-11	Balances that do not net to zero	Algorithm throws, no payments invented	Pass
TC-12	Second settlement while one is unpaid	409 SETTLEMENT_IN_PROGRESS	Pass
TC-13	Commit a receipt still processing	Refused; nothing written	Pass

ID	Test case	Expected result	Result
TC-14	Match "Basmati Rice 5kg" against 1 kg benchmark	No match (sizes differ)	Pass
TC-15	Monthly recurring template overdue by three months	Three entries created, schedule advanced	Pass
TC-16	Excel and PDF exports	Valid ZIP (XLSX) and file starting with %PDF	Pass
TC-17	Notifications collection unavailable	Expense creation still succeeds	Pass
TC-18	AED 240 expense at 1 AED = Rs 80	Stored as Rs 19,200 with the AED figures kept	Pass
TC-19	Three payers of AED 80 each at 1 AED = Rs 75.37	Payers 6,029 / 6,030 / 6,030, summing to the Rs 18,089 total	Pass
TC-20	Rate provider down, rates saved earlier	Saved rates returned and marked stale	Pass
TC-21	Rate provider down, no rates ever saved	503 FX_UNAVAILABLE	Pass
TC-22	Withdraw more than a goal holds	400 GOAL_BELOW_ZERO, goal unchanged	Pass
TC-23	Plain member records a settlement	403 NOT_ADMIN	Pass

6.4 Property-Based Testing of Settlement

The strongest evidence for the settlement engine comes from property-based tests. In each of 300 rounds, a random group of 2 to 10 members and 1 to 6 expenses is generated through the real split calculator; 100 further rounds use multi-payer expenses. For every round the test asserts three properties: (1) applying the returned payments brings every balance to exactly zero; (2) the number of payments is at most V − 1; and (3) every payment amount is a positive integer. All 400 rounds pass.

6.5 Security Testing

All 38 routes that existed at the time of the security review were audited for authentication, validation and ownership checks. Four gaps were found and fixed (Table 6.3). A dependency audit (`npm audit`) reported zero vulnerabilities after a transitive dependency was pinned. The rate limits were later raised (3,000 requests per 15 minutes overall and 60 sign-ins per 15 minutes per address) so that a whole classroom sharing one public IP address can use the system at once; bcrypt at cost 12 still makes password guessing slow.

Table 6.3: Security review findings and fixes

Gap	Risk	Fix
No global rate limit	Any authenticated endpoint could be flooded	Global limiter: 600 requests per 15 minutes
Unlimited receipt scans	Each scan calls a paid AI service	Scan limiter: 30 per hour

Gap	Risk	Fix
Permanent public image URLs	A leaked receipt link would work forever	Authenticated uploads; signed URLs valid for 1 hour
Proxy not trusted	All users behind the host proxy shared one rate-limit bucket	trust proxy enabled

6.6 Defects Found and Fixed

Table 6.4: Notable defects found during testing

Defect	Cause and fix
Phantom saving of Rs 1,470 on rice	Normalisation removed pack sizes, so 5 kg matched 1 kg. Fixed by parsing sizes separately and requiring them to agree.
Title-only edit wiped an expense split	Partial validation re-applied default values. Fixed by rebuilding missing split input from the stored splits.
Background price match clashed with user edits	Load-modify-save caused a version conflict. Fixed with an atomic update.
Password account reported as having no password	The password hash was excluded from the query. Fixed with a response transform instead.
Unmatched routes returned 401 instead of 404	Router-level authentication ran before the 404 handler. Fixed by applying authentication per route.
Personal page failed to render with a budget cap	A variable missed during a renaming refactor. Fixed and verified in the browser.
Blank page right after a deployment	Open tabs requested old, renamed code files. Fixed with an error boundary that reloads once when a code file is missing.
Offline OCR could stop the server	Tesseract.js downloaded language data at run time and a failed download crashed the process. Fixed by bundling the data and handling worker errors.
Invitation emails silently not delivered	The email provider refused messages until the account was activated, and mail errors were hidden. Fixed by logging every send and warning the admin, with WhatsApp and copy-link as alternatives.
Empty price book on a new database	Benchmark prices were only loaded by the seed script. Fixed by loading the default catalogue on start-up when the collection is empty.

6.7 Evaluation

Table 6.5 evaluates the system against its objectives.

Table 6.5: Evaluation against objectives

Objective	Evidence	Status
Personal workspace with budgets and recurring entries	FR-05 to FR-10 implemented; 16 personal tests pass	Achieved
AI receipt scanning with fallback and verification	FR-11 to FR-14; 18 receipt tests pass	Achieved

Objective	Evidence	Status
Savings suggestions from benchmark prices	FR-15; 100 products, 209 prices; 18 tests pass	Achieved
Group workspace with four exact split methods	FR-16 to FR-20; 22 calculator and 16 expense tests pass	Achieved
Settlement in at most V − 1 payments	400 property-test rounds; Murree Trip 20 → 4 payments	Achieved
Savings goals and multi-currency trips	FR-27 and FR-28; 31 currency, rate and goal tests pass; AED 412.50 = Rs 31,059 verified in the browser	Achieved
Reports and exports	FR-24 and FR-25; CSV, XLSX and PDF verified	Achieved
Secure authentication and authorisation	Token rotation, reuse detection, membership checks, audit	Achieved
Automated verification	288 of 288 tests passing	Achieved
Public, test-gated deployment	Live on Render with MongoDB Atlas; GitHub Actions must pass before each deploy	Achieved

CHAPTER 7
CONCLUSION AND FUTURE WORK

7.1 Conclusion

This project delivered SmartSplit & ExpenseAI, a secure web application that brings personal finance management and group expense settlement together in one account for Pakistani users. Receipt photos are converted into verified transactions with the help of a vision language model, verified items are compared with local market prices to suggest savings, and group debts are settled by a Min-Cash-Flow engine that never needs more than $V - 1$ payments. Savings goals help users plan ahead, trips abroad can be recorded in the currency on the bill, and the system is publicly deployed with every change tested before release.

Three design decisions proved central to the system's reliability: storing money as integer rupees with a single rounding rule, recomputing balances from the underlying expenses instead of storing them, and keeping a human in the loop for every AI-extracted value. Together with server-side authorisation and a comprehensive automated test suite of 288 passing tests, including property-based tests of the settlement guarantees, they give strong confidence that every rupee in the system is accounted for exactly.

7.2 Limitations

- The greedy settlement guarantees at most $V - 1$ payments but is not always the absolute minimum, which is an NP-hard problem.
- Price benchmarks come from a fixed local collection rather than live retailer data.
- The application records payments but does not transfer money itself.
- OCR accuracy depends on photo quality; poor images require more manual correction.
- A single typo in a short product name can fall below the 0.8 similarity threshold; common misspellings are handled through aliases instead.
- On free hosting tiers the server can still sleep when nobody uses it for a long time, so the first request after a pause is slow and the nightly jobs may run late.
- Exchange rates are refreshed once a day from a free provider, so a converted expense uses that day's market rate rather than the exact rate charged by the user's bank or card.
- Email delivery depends on the email provider; while sending is not available, invitations are shared by link or WhatsApp instead.

7.3 Future Work

1. Live price collection from retailer websites as a decoupled background job, with the local collection as a cache.
2. Integration with JazzCash, Easypaisa and Raast payment APIs so that a settlement payment can be made directly from the app.
3. A native mobile application reusing the existing REST API.
4. Automatic category suggestions using machine learning on the user's past transactions.
5. Automatic weekly import of official prices from the Pakistan Bureau of Statistics Sensitive Price Indicator (SPI) into the price book.
6. Letting the user enter the exact rate their card or bank charged for a foreign-currency expense.
7. An exact minimum-payment mode for small groups, using subset search, offered alongside the greedy method.

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APPENDIX A: API ENDPOINT REFERENCE

All endpoints are prefixed with `/api/v1`. Every endpoint except registration, sign-in, token refresh, Google sign-in, password reset and the health check requires a valid access token; group endpoints additionally require active membership, and admin endpoints the admin role.

Table A.1: REST API endpoints

Resource	Method	Path	Purpose
Auth	POST	`/auth/register`	Create an account
Auth	POST	`/auth/login`	Sign in with email and password
Auth	POST	`/auth/refresh`	Rotate refresh token, issue access token
Auth	POST	`/auth/logout`	Revoke the current session
Auth	GET	`/auth/me`	Current user
Auth	PATCH	`/auth/me`	Update profile
Auth	POST	`/auth/change-password`	Change password
Auth	POST	`/auth/google`	Google sign-in
Auth	POST	`/auth/forgot-password`	Email a reset link
Auth	POST	`/auth/reset-password/:token`	Set a new password
Users	GET	`/users/search`	Find users to add to a group
Groups	POST	`/groups`	Create a group
Groups	GET	`/groups`	List my groups with totals

Resource	Method	Path	Purpose
Groups	GET	`/groups/invites`	My invitations and join requests
Groups	POST	`/groups/:id/invites/accept`	Accept an invitation
Groups	DELETE	`/groups/:id/invites`	Decline an invitation or withdraw a request
Groups	PATCH	`/groups/:id/members/:memberId`	Promote or demote a member (admin)
Groups	GET	`/groups/:id`	Group details (member)
Groups	PATCH	`/groups/:id`	Update group (admin)
Groups	POST	`/groups/:id/members`	Add or invite a member (admin)
Groups	DELETE	`/groups/:id/members/:userId`	Remove a member (admin)
Groups	PATCH	`/groups/:id/members/:memberId/approve`	Approve a join request (admin)
Groups	POST	`/groups/:id/members/:memberId/resend`	Resend an invite (admin)
Groups	POST	`/groups/:id/join-code/rotate`	Rotate the share code (admin)
Groups	GET	`/join/:inviteToken`	Preview an invite link
Groups	POST	`/join/:inviteToken`	Accept an invite link
Groups	GET	`/join-request/:joinCode`	Preview a group by share code
Groups	POST	`/join-request/:joinCode`	Request to join

Resource	Method	Path	Purpose
Expenses	POST	`/groups/:id/expenses`	Add a group expense
Expenses	GET	`/groups/:id/expenses`	List expenses (filters, pages)
Expenses	GET	`/groups/:id/balances`	Net balances by aggregation
Expenses	GET	`/expenses/:expenseId`	One expense
Expenses	PATCH	`/expenses/:expenseId`	Edit an expense
Expenses	DELETE	`/expenses/:expenseId`	Soft-delete an expense
Settlements	POST	`/groups/:id/settle/preview`	Compute payments (no write)
Settlements	POST	`/groups/:id/settle`	Record a settlement (admin)
Settlements	GET	`/groups/:id/settlements`	Payments and audit trail
Settlements	PATCH	`/settlements/:settlementId/pay`	Mark a payment paid
Personal	POST	`/transactions`	Add a transaction
Personal	GET	`/transactions`	List transactions (filters)
Personal	PATCH	`/transactions/:transactionId`	Edit a transaction
Personal	DELETE	`/transactions/:transactionId`	Delete a transaction
Personal	GET	`/personal/summary`	Monthly summary
Personal	GET	`/budgets`	List budgets
Personal	PUT	`/budgets`	Create or update a budget

Resource	Method	Path	Purpose
Personal	DELETE	`/budgets/:budgetId`	Delete a budget
Goals	GET	`/goals`	List my goals with progress
Goals	POST	`/goals`	Create a goal
Goals	PATCH	`/goals/:goalId`	Edit a goal
Goals	DELETE	`/goals/:goalId`	Delete a goal
Goals	POST	`/goals/:goalId/contributions`	Deposit or withdraw
Goals	DELETE	`/goals/:goalId/contributions/:contributionId`	Remove a contribution
Categories	GET	`/categories`	Built-in and custom categories
Categories	POST	`/categories`	Add a custom category
Categories	GET	`/categories/:categoryId/usage`	Usage before deletion
Categories	DELETE	`/categories/:categoryId`	Delete a custom category
Receipts	POST	`/receipts/scan`	Upload a receipt (202)
Receipts	GET	`/receipts`	List receipts
Receipts	GET	`/receipts/:receiptId`	Receipt with OCR result
Receipts	POST	`/receipts/:receiptId/items`	Add a line
Receipts	PATCH	`/receipts/:receiptId/items/:index`	Edit a line
Receipts	DELETE	`/receipts/:receiptId/items/:index`	Remove a line
Receipts	POST	`/receipts/:receiptId/commit`	Save to personal or group draft
Reports	GET	`/reports/personal/monthly`	Monthly personal report

Resource	Method	Path	Purpose
Reports	GET	`/reports/personal/trends`	Six-month trends
Reports	GET	`/reports/groups/:id/summary`	Group summary report
Reports	GET	`/reports/export`	CSV, XLSX or PDF export
Notifications	GET	`/notifications`	Notification feed
Notifications	PATCH	`/notifications/:notificationId/read`	Mark one read
Notifications	PATCH	`/notifications/read-all`	Mark all read
Exchange rates	GET	`/fx/rates`	Rupee rates for supported currencies
Price book	GET	`/prices`	Benchmark prices by product and store
Price book	GET	`/prices/export`	Download as Excel
Price book	POST	`/prices`	Add or update a price (super admin)
Price book	DELETE	`/prices/:priceId`	Delete a price (super admin)
Price book	POST	`/prices/import`	Import Excel or CSV (super admin)
Price book	POST	`/prices/seed`	Load the default catalogue (super admin)
Admin	GET	`/admin/overview`	Totals and newest users and groups (super admin)

Resource	Method	Path	Purpose
System	GET	`/health`	Health and database status

APPENDIX B: USER MANUAL

This appendix explains how to perform the main tasks in the application.

Table B.1: Step-by-step guide to the main tasks

Task	Steps
Create an account	Open the website and select "Create an account", or use "Sign in with Google". Enter name, email and a password of at least 8 characters.
Record a transaction	Open Personal. In the form, choose Spent or Received, enter title, amount, date, category and payment method, optionally tick "Repeats" for a weekly or monthly bill, and select "Add transaction".
Set a budget	On the Personal page, open the budgets editor, choose a category and enter a monthly cap. Bars turn amber at 80% and red at 100%.
Scan a receipt	Open Scan and drop a photo or select "Choose image". When reading finishes, check every line in the dialog, correct anything wrong, then select "Save to Personal" or "Use in group expense".
Create a group	Open Groups and select "New group". Enter a name, type and start date, and add members by email.
Join a group	Open an invite link received by email or WhatsApp, accept an invitation from the inbox on the Groups page, or enter a share code and wait for an admin to approve the request.
Add a shared expense	In a group, select "Add expense". Enter title and amount, choose who paid (add more payers if needed), choose a split method and fill in the shares. Save becomes available when everything adds up.
Add an expense abroad	In Add expense, choose the currency (for example AED) next to the amount and type the amount from the bill. The form shows the rupee amount and the rate; the expense is saved in rupees.
Settle up	In a group, open Balances and select "Settle up", then "Simplify with Min-Cash-Flow". Review the payments; a group admin then selects "Record settlement". After paying through JazzCash, Easypaisa, Raast, bank or cash, select "Mark paid".
Save for a goal	Open Goals and select "New goal". Enter what you are saving for, the target amount and the date. Use Add or Withdraw on the goal card whenever you put money aside or take some out.
Reports and exports	Open Reports, choose the Personal or Group tab and the month, and use the CSV, Excel or PDF buttons to download.

APPENDIX C: INSTALLATION GUIDE

The following commands install and run the system locally. Node.js 20 or later and a MongoDB instance (local or Atlas) are required.

```
git clone <repository-url> smartsplit && cd smartsplit
npm run install:all
cp server/.env.example server/.env      # set MONGO_URI, JWT secrets, API keys
cp client/.env.example client/.env      # set VITE_API_URL
npm run seed                            # optional: demo data (erases the
database)
npm run dev                             # API on :5000, client on :5173
npm test                                # run the 288 automated tests
```

Required environment variables on the server are `MONGO_URI`, `JWT_ACCESS_SECRET`, `JWT_REFRESH_SECRET` and `CLIENT_URL`. Optional variables enable Google sign-in (`GOOGLE_CLIENT_ID`), the vision model (`ANTHROPIC_API_KEY`), Cloudinary (`CLOUDINARY_`*) and email (`SMTP_`*); without them the system falls back to Tesseract.js, inline images and console-printed email links, so it still runs completely offline.